

Grass

Solutions to Schröder

January 2023-Today

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Sequences of Real Numbers

2.5 Infinite Limits

Exercise 2.51. Prove **Cauchy's Limit Theorem**^a. That is, let $\{b_n\}_{n=1}^{\infty}$ be a strictly increasing sequence of positive numbers that go to infinity and let $\{a_n\}_{n=1}^{\infty}$ be a sequence. Prove that if the sequence $\left\{\frac{a_n - a_{n-1}}{b_n - b_{n-1}}\right\}_{n=2}^{\infty}$ converges to c , then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$.

Hint. Exercise 2-16 with $p_n := b_n - b_{n-1}$ and another appropriate sequence.

^aAlso known as the Stolz-Cesaro Theorem

Proof. Notice that $\lim_{n \rightarrow \infty} \sum_{k=1}^n b_{k+1} - b_k = \lim_{n \rightarrow \infty} b_{n+1} - b_1 = \infty$. Hence, by exercise 2-16,

$$c = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n (b_{k+1} - b_k) \left(\frac{a_{k+1} - a_k}{b_{k+1} - b_k} \right)}{\sum_{k=1}^n (b_{k+1} - b_k)} = \lim_{n \rightarrow \infty} \frac{a_{n+1} - a_1}{b_{n+1} - b_1} = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{b_{n+1} - b_1}.$$

Furthermore, we see that

$$\lim_{n \rightarrow \infty} \left(\frac{a_{n+1}}{b_{n+1} - b_1} - \frac{a_{n+1}}{b_{n+1}} \right) = \lim_{n \rightarrow \infty} b_1 \cdot \lim_{n \rightarrow \infty} \frac{a_{n+1}}{b_{n+1} - b_1} \cdot \lim_{n \rightarrow \infty} \frac{1}{b_{n+1}} = 0.$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c.$$



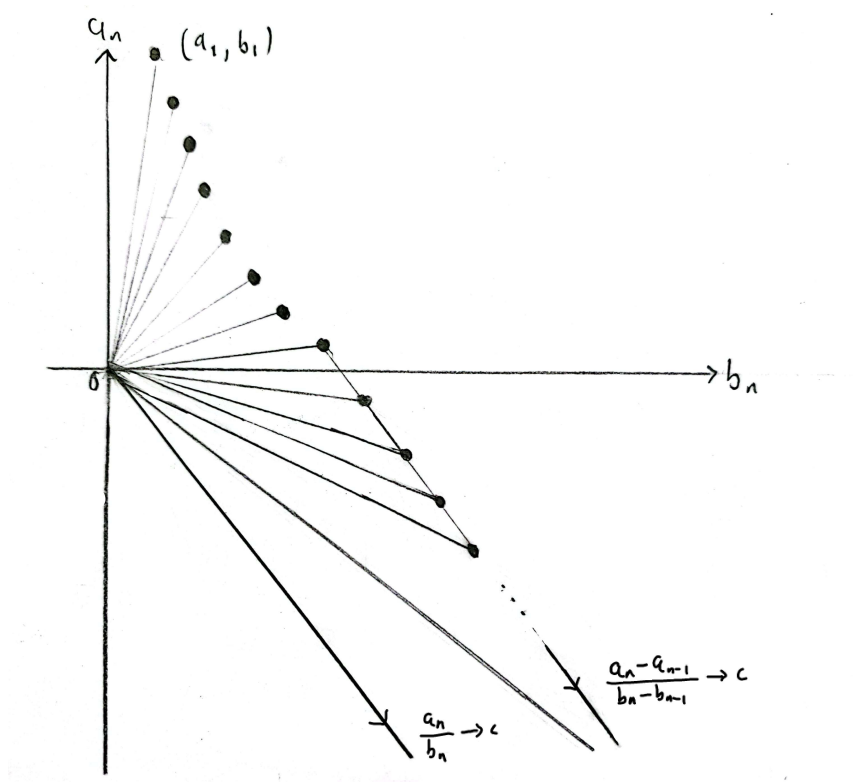


Figure 2.1: An Illustration of Cauchy's Limit Theorem

Continuous Functions

3.5 Properties of Continuous Functions

Theorem 3.34. The Intermediate Value Theorem Let $a < b$ and let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. If $f(a) < 0$ and $f(b) > 0$ (or vice versa) then there is a $c \in (a, b)$ such that $f(c) = 0$ (Also see Figure 8(a).)

V1:

Proof. Assume, for the sake of contradiction, that $f(c) \neq 0$ for all $c \in (a, b)$.

First consider the case where for all $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ so $0 < f(x_n) < 1/n$. By AC and the Bolzano-Weierstrass Theorem, we can construct a sequence $\{x_n\}_{n=1}^{\infty}$ of such x_n , that converges to some limit L and whose image under f , $\{f(x_n)\}_{n=1}^{\infty}$, goes to 0. Consequently, from The Squeeze Theorem and continuity, $f(L) = \lim_{n \rightarrow \infty} f(x_n) = 0$, contradicting our initial assumption that $0 \notin f([a, b])$.

Now consider when there exists $\varepsilon > 0$ so for any y , if $0 < y < \varepsilon$ then $y \notin f[a, b]$. Let $\varepsilon_s > 0$ be the supremum of such ε . For each $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ with $\varepsilon_s \leq f(x_n) < \varepsilon_s + 1/n$. Again, construct a sequence $\{x_n\}_{n=1}^{\infty}$ of such x_n , which converges to some limit L . Its image under f , $\{f(x_n)\}_{n=1}^{\infty}$, must also converge to ε_s . Let L_i be the infimum of such limits L . For every $m \in \mathbb{N}$, there is some limit L_m (possibly L_i itself) for which $L_m - L_i < 1/m$. Thus, also a corresponding sequence $\{x_{m,n}\}_{n=1}^{\infty}$ converging to L_m ; such that for any $n \in \mathbb{N}$, there is a $k_n \in \mathbb{N}$ with $\varepsilon_s \leq f(x_{m,k_n}) < \varepsilon_s + 1/n$. By having $\bar{x}_m$ be some x_{m,k_m} , we can once more construct a sequence $\{\bar{x}_m\}_{m=1}^{\infty}$ converging to L_i , so $\lim_{z \rightarrow L_i} f(z) = \lim_{n \rightarrow \infty} f(\bar{x}_n) = \varepsilon_s$ by continuity.

If there is some $\delta > 0$ for which every z with $L_i - \delta < z < L_i$ has $f(z) \geq \varepsilon_s$, then let δ_s be the supremum of such δ . Then either $L_i - \delta_s = a$, hence $0 < \lim_{z \rightarrow a^+} f(z) =$

$f(a) < 0$, or if $L_i - \delta_s < a$, then $\lim_{z \rightarrow (L_i - \delta_s)^-} f(z) < 0$ but $\lim_{z \rightarrow (L_i + \delta_s)^+} f(z) > 0$. A contradiction to continuity. Accordingly, no such $\delta > 0$ can exist. Instead, this tells us that there exists a sequence $\{y_n\}_{n=1}^\infty$ in $[a, L_i)$ converging to L_i , having $y_n < L_i$ and $f(y_n) < 0$, for all $n \in \mathbb{N}$. Thence, $\lim_{z \rightarrow L_i} f(z) = \lim_{n \rightarrow \infty} f(y_n) \leq 0$ by continuity, a contradiction to the conclusion in the prior paragraph. 

V2:

Proof. Assume, for the sake of contradiction, that $f(c) \neq 0$ for all $c \in (a, b)$.

First consider the case where for all $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ so $f(x_n) \in (0, 1/n)$. By AC and the Bolzano-Weierstrass Theorem, there is hence a sequence $\{x_n\}_{n=1}^\infty$ converging to some $L \in [a, b]$ having $f(x_n) \in (0, 1/n)$ for each n , so continuity with The Squeeze Theorem says $f(L) = \lim_{n \rightarrow \infty} f(x_n) = 0$. A contradiction.

Now consider when there exists $\varepsilon > 0$ so for any y , if $y \in (0, \varepsilon)$ then $y \notin f[a, b]$. Let $\varepsilon_s > 0$ be the supremum of all such ε , bounded above by $f(b)$; allowing us to again construct a sequence $\{x_n\}_{n=1}^\infty$ converging to some $L \in [a, b]$, but with $f(x_n) \in [\varepsilon_s, \varepsilon_s + 1/n)$ for each $n \in \mathbb{N}$.

Let L_i be the infimum of all possible limits $L \in [a, b]$, for which there exists any accompanying sequence $\{x_n\}_{n=1}^\infty$ as described above which converges to L . For every $m \in \mathbb{N}$, there therefore is a limit $L_m \in [L_i, L_i + 1/m)$ and its corresponding sequence $\{x_{m,n}\}_{n=1}^\infty$ with some $x_{m,k_m} \in (L_m - 1/m, L_m + 1/m) \subseteq (L_i - 1/m, L_i + 2/m)$ and $f(x_{m,k_m}) \in [\varepsilon_s, \varepsilon_s + 1/m)$. Choosing a x_{m,k_m} for each $\bar{x}_m$, we have a sequence $\{\bar{x}_m\}_{m=1}^\infty$ that converges to L_i , such that $\lim_{z \rightarrow L_i} f(z) = \varepsilon_s$.

Finally, let $y_i := \inf\{x \mid f(x) \geq \varepsilon_s\}$ and presume $y_i \in (a, L_i)$. Thus $f(y_i) = \lim_{z \rightarrow y_i^-} f(z) \leq 0$ and it must be that, for any $n \in \mathbb{N}$, there exists $y_n \in (y_i, y_i + 1/n)$ so $f(y_n) \geq \varepsilon_s$; $\lim_{z \rightarrow y_i^+} f(z) = \lim_{n \rightarrow \infty} f(y_n) \geq \varepsilon_s$. A contradiction, implying $y_i > L_i$. However, now $\lim_{z \rightarrow L_i^-} f(z) \leq 0$, contradicting $\lim_{z \rightarrow L_i} f(z) = \varepsilon_s$.

Wherefore, our initial assumption is impossible, suggesting that the Intermediate Value Theorem is true. 

Differentiable Functions

4.3 Rolle's Theorem and the Mean Value Theorem

Exercise 4.24. Let $f: (a, b) \rightarrow \mathbb{R}$ be continuous and let $x \in (a, b)$. Prove that if $f'(z)$ exists for all $z \in (a, b) \setminus \{x\}$ and $\lim_{z \rightarrow x} f'(z)$ exists, then f is differentiable at x with $f'(x) = \lim_{z \rightarrow x} f'(z)$.

Proof. Let $\varepsilon > 0$, we shall unpack 3 definitions first:

- (a) Since $L := \lim_{z \rightarrow x} f'(z)$ exists, there be some $\delta_1 > 0$ such that whenever $y \in (a, b) \setminus \{x\}$ has $|y - x| < \delta_1$, then $\left| \lim_{z \rightarrow y} \frac{f(z) - f(y)}{z - y} - L \right| < \frac{1}{4}\varepsilon$.
- (b) Given any $y \in (a, b)$, we have the limit $L_y := \lim_{z \rightarrow y} \frac{f(z) - f(y)}{z - y}$. So for some $\delta_2 > 0$, if $z \in (a, b) \setminus \{y\}$ and $|z - y| < \delta_2$, then $\left| \frac{f(z) - f(y)}{z - y} - L_y \right| < \frac{1}{4}\varepsilon$.
- (c) Finally, by limit laws, there is a $\delta_3 > 0$, so for each $y \in (a, b) \setminus \{x\}$ and $|y - x| < \delta_3$, $\left| \frac{f(z) - f(x)}{z - x} - L \right| < \frac{1}{2}\varepsilon$.

Now, define $\delta := \min\{\delta_1, \delta_2/2, \delta_3\}$ and let $z \in (a, b) \setminus \{x\}$ so $|z - x| < \delta$, and $y \in (a, b) \setminus \{x\}$ with $|y - x| < \delta$. Then, $|z - y| - |y - x| \leq |z - y - (y - x)| = |z - x| < \delta$. Accordingly, $|z - y| < |y - x| + \delta < \delta_2/2 + \delta_2/2 = \delta_2$. Therefore, the inequalities of the previous paragraph hold. Which means,

$$\left| \frac{f(z) - f(y)}{z - y} - L \right| = \left| \frac{f(z) - f(y)}{z - y} - L_y + (L_y - L) \right| < \frac{1}{2}\varepsilon.$$

As such,

$$\left| \frac{f(z) - f(x)}{z - x} - L \right| \leq \left| \frac{f(z) - f(x)}{z - x} - \frac{f(z) - f(y)}{z - y} \right| + \left| \frac{f(z) - f(y)}{z - y} - L \right| < \frac{1}{2}\varepsilon + \frac{1}{2}\varepsilon.$$

In other words, we have shown that for any $\varepsilon > 0$, there is $\delta > 0$, so when $z \in (a, b) \setminus \{x\}$ such that $|z - x| < \delta$, $\left| \frac{f(z) - f(x)}{z - x} - L \right| < \varepsilon$ holds true. Consequently, we have shown that $f'(x) := \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$ exists, and is in fact just $L :=$

$$\lim_{z \rightarrow x} f'(x).$$



Exercise 4.25 (Darboux's theorem). Let $f: (a, b) \rightarrow \mathbb{R}$ be differentiable. Prove that f' (which need not be continuous) has the *intermediate value property*. That is, prove that for all $c < d$ in (a, b) and all v between $f'(c)$ and $f'(d)$ there is an $m \in (c, d)$ so that $f'(m) = v$.

Proof. (*Work in progress*)



Exercise. Is f' continuous a.e.?

The Riemann Integral I

5.2 Uniform Continuity and Integrability of Continuous Functions

Exercise 5.1. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded and $g: [a, b] \rightarrow \mathbb{R}$ be nondecreasing. With m_i and M_i as in Definition 5.13, let $L_g(f, P) := \sum_{i=1}^n m_i \Delta g_i$ be the *lower sum* and let $U_g(f, P) := \sum_{i=1}^n M_i \Delta g_i$ be the *upper sum* of f with respect to g and P .

- (a) Let P, Q be partitions of $[a, b]$ so that Q is a refinement of P . Prove that $L_g(f, P) \leq L_g(f, Q) \leq U_g(f, Q) \leq U_g(f, P)$.
- (b) Prove that if f is continuous, then f is Riemann-Stieltjes integrable with respect to g .
- (c) Let $f_1, f_2: [a, b] \rightarrow \mathbb{R}$ be Riemann-Stieltjes integrable with respect to g . Prove that if $f_1 \leq f_2$, then $\int_a^b f_1 dg \leq \int_a^b f_2 dg$.
- (d) *Mean Value Theorem for Riemann Stieltjes Integrals.* Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Prove that there is a $c \in (a, b)$ so that $\int_a^b f dg = f(c)(g(b) - g(a))$.
- (e) Let $c \in (a, b)$ and let $g := \mathbf{1}_{[c, b]}$.
 - i. Prove that if $f: [a, b] \rightarrow \mathbb{R}$ is continuous at c , then f is Riemann-Stieltjes integrable with respect to g and $\int_a^b f d\mathbf{1}_{[c, b]} = f(c)$.
 - ii. Prove that if $f: [a, b] \rightarrow \mathbb{R}$ is not continuous at c , then f is not Riemann-Stieltjes integrable with respect to g .
- (f) Prove that if f is continuous and g is continuously differentiable, then, with the integral on the right being a Riemann integral, we have $\int_a^b f dg = \int_a^b f(x)g'(x) dx$.

Note. Integrals against dg are often used to abbreviate Riemann integrals as on the right side.

Proof. (a) Let $Q := \{x_i\}_0^N$ and $P := \{x_{i_k}\}_0^n$. Since $m_{i_k} \leq m_i$ and $M_{i_k} \geq M_i$ for $i_k \leq i \leq i_{k+1}$, (where $m_{i_k} := \inf f[x_{i_{k-1}}, x_{i_k}]$ and $\Delta g_{i_k} := g(x_{i_k}) - g(x_{i_{k-1}})$)

$$m_{i_k} \Delta g_{i_k} = \sum_{i=i_k}^{i_{k+1}} m_{i_k} \Delta g_i \leq \sum_{i=i_k}^{i_{k+1}} m_i \Delta g_i.$$

So, $L_g(f, P) \leq L_g(f, Q)$. Similarly, $U_g(f, Q) \leq U_g(f, P)$.

(b) If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, it is uniformly continuous. So, for $\varepsilon > 0$, pick $\delta > 0$ such that

$$\mathcal{L} - L_g(f, P) < \frac{\varepsilon}{2} \quad \text{and} \quad |f(x) - f(y)| < \frac{\varepsilon}{2[g(b) - g(a)]}.$$

Now, for any partition P and evaluation set $T = \{t_i\}_0^n$,

$$\begin{aligned} |R_g(f, P, T) - \mathcal{L}| &\leq R_g(f, P, T) - L_g(f, P) + \mathcal{L} - L_g(f, P) \\ &< \sum [f(t_i) - m_i] \Delta g_i + \varepsilon/2 \\ &\leq \varepsilon. \end{aligned}$$

(c) Since $f_2 - f_1 \geq 0$, it is clear that $R_g(f_2 - f_1, P, T) \geq 0$ for all partitions P and evaluation sets T . Hence, $\int_a^b f_2 - f_1 dg \geq 0$.

(d) Choose $f(m) = \min f$ and $f(M) = \max f$. Then,

$$f(m) \leq \frac{\int_a^b f dg}{g(b) - g(a)} \leq f(M)$$

so the Intermediate Value Theorem applies.

(e) Previously completed on paper.

(f) Let $\varepsilon > 0$ and $T := \{t_i\}_0^n$ be an evaluation set. Pick $\delta > 0$ such that, for $M := \max f$,

$$\left| \frac{g(x) - g(t)}{x - t} - g'(x) \right| < \frac{\varepsilon}{3M(b-a)}$$

if $|t - x| < \delta$. Select a partition P with $\|P\| < \delta$, such that

$$\left| \int_a^b f dg - R_g(f, P, T) \right| < \frac{\varepsilon}{3} \quad \text{and} \quad \left| \int_a^b f(x)g'(x) dx - R(f, P, T) \right| < \frac{\varepsilon}{3}.$$

Notice that

$$\begin{aligned} |R_g(f, P, T) - R(f, P, T)| &= \left| \sum f(t_i) [\Delta g_i - g'(x_i) \Delta x_i] \right| \\ &< \sum M \cdot \frac{\varepsilon \Delta x_i}{3M(b-a)} \\ &= \frac{\varepsilon}{3}. \end{aligned}$$

Hence,

$$\left| \int_a^b f dg - \int_a^b f(x)g'(x) dx \right| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$



5.3 The Fundamental Theorem of Calculus

Exercise 5.20. Power Rule for integration. Let $r \in \mathbb{Q} \setminus \{1\}$ and let $a < b$. In case $r < 0$, let a and b either both be positive or both be negative. Prove that

$$\int_a^b x^r dx = \frac{1}{r+1} b^{r+1} - \frac{1}{r+1} a^{r+1}.$$

Then explain why we needed to require a and b to be both positive or both negative for $r < 0$.

Proof. This is clear from FTC and the Power Rule for differentiation. For $r < 0$, notice 0^r is undefined. Furthermore, even if we were to modify the integrand to be defined at $x = 0$, we see that x^r would still be unbounded on $[a, b]$. So, 5-10 tells us x^r is not integrable on $[a, b]$.



Exercise 5.21. Integration by Parts. Let $[a, b] \subset (c, d)$ and let $F, g: (c, d) \rightarrow \mathbb{R}$ be continuously differentiable with derivatives f and g' . Prove that

$$\int_a^b f(x)g(x) dx = F(b)g(b) - F(a)g(a) - \int_a^b F(x)g'(x) dx.$$

Proof. By the Product Rule,

$$(Fg)' = fg + Fg'.$$

Since products of continuous functions are continuous, they are Riemann integrable. Hence FTC implies

$$\int_a^b f(x)g(x) dx = F(b)g(b) - F(a)g(a) - \int_a^b F(x)g'(x) dx.$$



Exercise 5.22. Integration by Substitution. Let $[a, b] \subset (c, d)$, let $g: (c, d) \rightarrow \mathbb{R}$ be continuously differentiable with derivative g' and let F be continuously differentiable with derivative f such that the domain of F contains $g[[a, b]]$. Prove that

$$\int_a^b f(g(x))g'(x) dx = F(g(b)) - F(g(a)).$$

Proof. By the Chain Rule,

$$(F \circ g)'(x) = f(g(x))g'(x).$$

Since products/compositions of continuous functions are continuous by limit laws/Theorem 3.30, they are Riemann integrable. By FTC,

$$\int_a^b f(g(x))g'(x) dx = F(g(b)) - F(g(a)).$$



5.4 The Darboux Integral

Theorem 5.29. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then f is Darboux integrable on $[a, b]$ iff f is Riemann integrable on $[a, b]$. Moreover, the Darboux and Riemann integrals are equal in this case, that is,

$$\int_a^b f(x) dx = \mathcal{L} = \mathcal{U}.$$

Proof. Let $\varepsilon > 0$.

First consider f being Darboux integrable. From Lemma 5.16, there must be a partition R of $[a, b]$ so $\mathcal{L}_f - L(f, R) < \frac{1}{2}\varepsilon$ and $U(f, R) - \mathcal{U}_f < \frac{1}{2}\varepsilon$. Hence, f is Riemann integrable since

$$U(f, R) - L(f, R) \leq U(f, R) - \mathcal{L}_f + \mathcal{L}_f - L(f, R) < \varepsilon.$$

Conversely, when f is Riemann integrable, there is a partition S with $\mathcal{U}_f - L(f, S) < \frac{1}{2}\varepsilon$ and $U(f, S) - \mathcal{L}_f < \frac{1}{2}\varepsilon$. Consequently, f is Darboux integrable as

$$\mathcal{U}_f - \mathcal{L}_f \leq \mathcal{U}_f - L(f, S) - \mathcal{L}_f + U(f, S) < \varepsilon.$$

Furthermore, Lemma 5.14 guarantees the equality

$$\int_a^b f(x) dx = \mathcal{L} = \mathcal{U}.$$



Theorem. If $g: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, so is $|g|$.

Proof. Let $\varepsilon > 0$. Define $n_i := \min\{|m_i|, |M_i|\}$ and $N_i = |m_i| + |M_i| - n_i$. Wlog, $n_i = |m_i|$. Notice that

$$N_i - n_i = |M_i| - n_i + |m_i| - n_i \leq |M_i - m_i|.$$

By Riemann's Condition, there is a partition P for which $U(g, P) - L(g, P) < \varepsilon$. As such,

$$U(|g|, P) - L(|g|, P) = \sum (N_i - n_i) \Delta x_i \leq \sum (M_i - m_i) \Delta x_i < \varepsilon.$$



Lemma. Let $g: [a, b] \rightarrow \mathbb{R}$ be Riemann integrable. Then, for any sequence of partitions $\{P_k\}_{k=1}^{\infty}$ with norm $\|P_k\|$ converging to 0,

$$\lim_{k \rightarrow \infty} L(g, P_k) = \int_a^b g(x) dx.$$

Proof. Let $\varepsilon > 0$. For each k , there is t_i such that

$$g(t_i) - m_i < \frac{\varepsilon}{2k\Delta x_i}.$$

Accordingly, for the evaluation set T_k consisting of all such t_i ,

$$R(g, P_k, T_k) - L(g, P_k) < \frac{1}{2}\varepsilon.$$

By Lemma 5.6, there is some K , such that when $k \geq K$, we have

$$\left| R(g, P_k, T_k) - \int_a^b g(x) dx \right| < \frac{1}{2}\varepsilon.$$

Hence, using the triangular inequality,

$$\left| L(g, P_k) - \int_a^b g(x) dx \right| < \varepsilon.$$



Exercise 5.26. Let $f: [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$.

- (a) For each $n \in \mathbb{N}$, let $P_n = \{a = x_0^{(n)} < \dots < x_{k_n}^{(n)} = b\}$ be a partition of the interval $[a, b]$ with $\|P_n\| < \frac{1}{n}$ and let

$$s_n := \left(\sum_{i=1}^{k_n-1} m_i^{(n)} \mathbf{1}_{[x_{i-1}^{(n)}, x_i^{(n)})} \right) + m_{k_n}^{(n)} \mathbf{1}_{[x_{k_n-1}^{(n)}, x_{k_n}^{(n)}]},$$

where $m_i^{(n)} = \inf \{f(x) \mid x \in [x_{i-1}^{(n)}, x_i^{(n)}]\}$.

- Prove that for all $n \in \mathbb{N}$ and all $x \in [a, b]$ we have $s_n(x) \leq f(x)$.
- Prove that if f is continuous at $x \in [a, b]$, then $\{s_n(x)\}_{n=1}^{\infty}$ converges to $f(x)$.

iii. Prove that

$$\lim_{n \rightarrow \infty} \int_a^b |f - s_n| dx = 0.$$

(b) Prove that there is a sequence $\{c_n\}_{n=1}^\infty$ of continuous functions on $[a, b]$ such that for all $n \in \mathbb{N}$ we have $|c_n| \leq |f|$ and so that

$$\lim_{n \rightarrow \infty} \int_a^b |f - c_n| dx = 0.$$

Proof.

(a)

i. Fix $n \in \mathbb{N}$ and $x \in [a, b]$. There exists i with $x \in [x_{i-1}^{(n)}, x_i^{(n)})$. So,

$$s_n(x) = m_i^{(n)} \leq f(x).$$

ii. Let f be continuous at $x \in [a, b]$, and $\varepsilon > 0$. Then, there is $\delta > 0$, such that if $|z - x| < \delta$, then $|f(z) - f(x)| < \varepsilon/2$. Moreover, $\delta > 1/K$ for some $K \in \mathbb{N}$. When $k \geq K$, there also exists $|y - x| < 1/k$ with $|f(y) - s_k(x)| < \varepsilon/2$. Consequently,

$$f(x) - s_k(x) \leq |f(x) - f(y)| + |f(y) - s_k(x)| < \varepsilon.$$

iii. Let $\varepsilon > 0$. By our lemma, there exists $N \in \mathbb{N}$ for which when $n \geq N$,

$$\int_a^b |f - s_n| dx = \int_a^b f(x) dx - \int_a^b s_n(x) dx = \int_a^b f(x) dx - L(f, P_n) < \varepsilon.$$

(b) Let $c_n, S_n: [a, b] \rightarrow \mathbb{R}$ be defined by

$$c_n(x) := \frac{f(x_{m+1}^{(n)}) - f(x_m^{(n)})}{x_{m+1}^{(n)} - x_m^{(n)}} (x - x_m^{(n)}) + f(x_m^{(n)}) \quad \text{for } x \in [x_m^{(n)}, x_{m+1}^{(n)}]$$

$$S_n(x) := \left(\sum_{i=1}^{k_n-1} M_i^{(n)} \mathbf{1}_{[x_{i-1}^{(n)}, x_i^{(n)})} \right) + M_{k_n}^{(n)} \mathbf{1}_{[x_{k_n-1}^{(n)}, x_{k_n}^{(n)}]}.$$

The continuity of c_n is clear since it is made of line segments joined together. Since $c_n(x), f(x) \in [s_n(x), S_n(x)]$,

$$\int_a^b |f - c_n| dx \leq \int_a^b S_n - s_n dx = U(f, P_n) - L(f, P_n) < \varepsilon$$

for sufficiently large n .



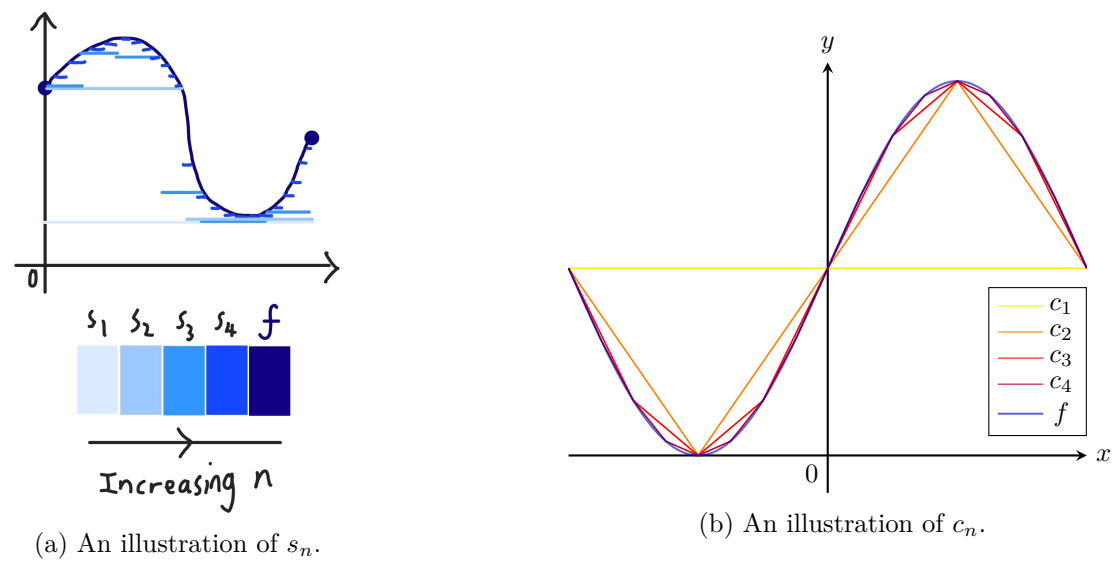


Figure 5.1

Exercise 5.27. Prove **Riemann's Condition** for Riemann-Stieltjes integrals. That is, let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. Let $g: [a, b] \rightarrow \mathbb{R}$ be nondecreasing and prove that f is Riemann-Stieltjes integrable on $[a, b]$ with respect to g iff for all $\varepsilon > 0$ there is a partition P of $[a, b]$ such that $U_g(f, P) - L_g(f, P) < \varepsilon$.

Series of Real Numbers I

6.1 Series as a Vehicle To Define Infinite Sums

Definition. Let $\{x_n\}_{n=1}^{\infty}$ be sequence converging to 0, and m_n to be the least number such that $|x_{m_n} - L| < 1/n$. The *1-rate of convergence* and *2-rate of convergence* are, respectively,

$$r_1 := \lim_{n \rightarrow \infty} \frac{m_{n+1}}{m_n} \quad \text{and} \quad r_2 := \lim_{n \rightarrow \infty} \frac{m_{n+2} - m_{n+1}}{m_{n+1} - m_n}.$$

Question. Does r_2 always exist?

Question. Is there any range of values of r_1 and/or r_2 for which $\sum_{n=1}^{\infty} x_n$ is guaranteed to converge? Can we establish a biconditional?

Exercise 6.2. Compute the value of each of the series below

(d)

$$\sum_{j=4}^{\infty} \frac{2^{j-2}}{5^{j+4}}$$

Proof. (d) By theorem 6.2, we compute

$$\sum_{j=4}^{\infty} \frac{2^{j-2}}{5^{j+4}} = \frac{1}{2500} \cdot \frac{2}{5} \left[\frac{1}{1 - \frac{2}{5}} - \frac{1 - \left(\frac{2}{5}\right)^3}{1 - \frac{2}{5}} \right] = \frac{4}{234375}.$$



Exercise 6.3. More on the arithmetic of series (Theorem 6.4).

- (c) Give an example of two series $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ such that $\sum_{j=1}^{\infty} a_j + b_j$ converges but neither $\sum_{j=1}^{\infty} a_j$ nor $\sum_{j=1}^{\infty} b_j$ converges.
- (d) Is it possible to find two series $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ such that $\sum_{j=1}^{\infty} a_j + b_j$ converges, and exactly one of $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ converges.
- (e) Is there a series $\sum_{j=1}^{\infty} a_j$ and a $c \in \mathbb{R}$ such that $\sum_{j=1}^{\infty} ca_j$ converges, but $\sum_{j=1}^{\infty} a_j$ diverges?

Proof. (c) Let $a_j = 1$ and $b_j = -1$ for all j . Then, $\sum_{j=1}^{\infty} a_j = \infty$ and $\sum_{j=1}^{\infty} b_j = -\infty$, while $\sum_{j=1}^{\infty} a_j + b_j = 0$.

- (d) It is impossible. Let $\sum_{j=1}^{\infty} a_j = S$ be a convergent series, and $\sum_{j=1}^{\infty} b_j$ a divergent series.

Also let $L \in \mathbb{R}$. Then, there exists $\varepsilon > 0$ and some $n \in \mathbb{N}$, for which $\left| \sum_{j=1}^{\infty} a_j - S \right| < \varepsilon$ and $\left| \sum_{j=1}^{\infty} b_j - (L - S) \right| \geq 2\varepsilon$. Then, by the reverse triangle inequality,

$$\left| \sum_{j=1}^{\infty} a_j + b_j - L \right| \geq \left| \sum_{j=1}^{\infty} b_j - (L - S) \right| - \left| \sum_{j=1}^{\infty} a_j - S \right| \geq 2\varepsilon - \varepsilon = \varepsilon.$$

- (e) For any divergent series $\sum_{j=1}^{\infty} a_j$, such as the harmonic series, $\sum_{j=1}^{\infty} 0 \cdot a_j = 0$. But if $c \neq 0$, then $\sum_{j=1}^{\infty} ca_j$ clearly must diverge.



Lemma. Let $\{d_j\}_{j=1}^n$ be a sequence of digits from 0 to 9. Then,

$$0.\overline{d_1 d_2 \dots d_n} = \frac{d_1 d_2 \dots d_n}{\underbrace{99 \dots 9}_{n \text{ times}}}.$$

Proof. It is clear that

$$0.\overline{d_1 d_2 \dots d_n} = d_1 d_2 \dots d_n \sum_{j=1}^{\infty} 10^{-n} = \frac{d_1 d_2 \dots d_n \cdot 10^{-n}}{1 - 10^{-n}} = \frac{d_1 d_2 \dots d_n}{\underbrace{99 \dots 9}_{n \text{ times}}}.$$



Exercise 6.4. Convert each of the following infinite repeating decimals below into a fraction. A bar over a set of digits means that these digits repeat indefinitely.

- (a) 0.25, (b) $0.\overline{25}$, (c) $0.\overline{9462}$, (d) $0.\overline{1473}$, (e) $12.0049\overline{5}$.

Proof. By the above lemma, we see that

$$\begin{aligned} \text{(a)} \quad 0.25 &= \frac{1}{4}, & \text{(b)} \quad 0.\overline{25} &= \frac{25}{99}, & \text{(c)} \quad 0.\overline{9462} &= \frac{9462}{9999} = \frac{3153}{3333}, \\ \text{(d)} \quad 0.\overline{1473} &= \frac{1473}{9999}, & \text{(e)} \quad 12.0049\overline{5} &= 12 + \frac{4}{1000} + \frac{95}{99} \cdot 10^{-3} = \frac{1188491}{99000}. \end{aligned}$$



Exercise 6.8. 2^k test. Let $\sum_{j=1}^{\infty} a_j$ be a nonincreasing sequence with nonnegative terms. Prove that $\sum_{j=1}^{\infty} a_j$ converges iff $\sum_{j=1}^{\infty} 2^k a_{2^k}$ converges.

Hint. Use Example 6.8 as guidance.

Proof. Assume $\sum_{k=1}^{\infty} 2^k a_{2^k}$ diverges. By nonnegativity, $\sum_{j=1}^{\infty} 2^k a_{2^k} = \infty$ is clear. Hence,

$$\sum_{j=1}^{2^n} a_j = a_1 + \sum_{k=0}^{n-1} \sum_{j=2^k+1}^{2^{k+1}} a_j \geq a_1 + \sum_{k=0}^{n-1} 2^k a_{2^{k+1}} \geq \frac{1}{2} a_1 + \frac{1}{2} \sum_{k=0}^n 2^k a_{2^k}.$$

So, $\sum_{j=1}^{\infty} a_j = \infty$.

Now suppose $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges to $S \geq 0$. Similarly,

$$\sum_{j=1}^{2^n} a_j = a_1 + \sum_{k=0}^{n-1} \sum_{j=2^k+1}^{2^{k+1}} a_j \leq a_1 + \sum_{k=0}^{n-1} 2^k a_{2^k} \leq a_1 + S.$$

Since $\sum_{j=1}^n a_j$ is nondecreasing and bounded, it definitely converges. 

Exercise 6.9. Prove that $\sum_{j=1}^{\infty} (-1)^{j+1} \frac{1}{j}$ converges by showing that the partial sums form a Cauchy sequence.

Proof. Let $\varepsilon > 0$ and pick $N \in \mathbb{N}$, such that $\varepsilon > N^{-1}$. Then, for every $m \geq n \geq N$ we have

$$\left| \sum_{j=n+1}^m (-1)^{j+1} \frac{1}{j} \right| = \left| \sum_{j=1}^{m-n} (-1)^{n+1+j} \frac{1}{n+j} \right| \leq \sum_{\substack{j=1 \\ j \text{ odd}}}^{m-n} \frac{1}{n+j} - \sum_{\substack{j=1 \\ j \text{ even}}}^{m-n} \frac{1}{n+j+1} \leq \frac{1}{n+1}$$

which is less than ε . 

Exercise 6.10. Translating between sequences and series. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. Prove that $\{a_n\}_{n=1}^{\infty}$ converges iff the series $\sum_{j=1}^{\infty} a_{j+1} - a_j$ converges and that in this case we obtain the limit as $\lim_{n \rightarrow \infty} a_n = a_1 + \sum_{j=1}^{\infty} a_{j+1} - a_j$.

Proof. This is apparent by noticing that $\sum_{j=1}^n a_{j+1} - a_j = a_{n+1} - a_1$, for any $n \in \mathbb{N}$ and sequence $\{a_n\}_{n=1}^{\infty}$. 

Exercise 6.11. Use Lemma 6.3 to prove the Monotone Sequence Theorem.

Proof. Wlog, let $\{a_n\}_{n=1}^{\infty}$ be a nondecreasing sequence bounded above by u . So, $a_{j+1} - a_j \geq 0$ for each j , and $\{\sum_{j=1}^n a_{j+1} - a_j\}_{n=1}^{\infty}$ is bounded above by $u + |a_1|$. Hence, Lemma 6.3 tells us $\sum_{j=1}^{\infty} a_{j+1} - a_j$ converges. The preceding exercise implies $\{a_n\}_{n=1}^{\infty}$ converges. 

Exercise 6.12. Let $\sum_{j=1}^{\infty} a_j$ be a convergent series, let $\{j_k\}_{k=0}^{\infty}$ be a strictly increasing sequence of natural numbers with $j_0 = 1$ and for all $k \in \mathbb{N}$ let $A_k := \sum_{j=j_{k-1}}^{j_k-1} a_j$. Prove that $\sum_{k=1}^{\infty} A_k$ converges and $\sum_{k=1}^{\infty} A_k = \sum_{j=1}^{\infty} a_j$.

Proof. We see that $\sum_{k=1}^n A_k = \sum_{j=1}^{j_n-1} a_j$ for each $n \in \mathbb{N}$. Hence, the result follows immediately. (Recall that for any convergent sequence $\{s_n\}_{n=1}^{\infty}$, all subsequences of $\{s_n\}_{n=1}^{\infty}$ have the same limit as $\{s_n\}_{n=1}^{\infty}$.) 

6.2 Absolute Convergence and Unconditional Convergence

Theorem 6.11 (Alternating Series Test). Let $\{b_j\}_{j=1}^{\infty}$ be a nonincreasing nonnegative sequence such that $\lim_{j \rightarrow \infty} b_j = 0$. Then, $\sum_{j=1}^{\infty} (-1)^{j+1} b_j$ converges.

Proof. This is clear from imitating what was done for exercise 6.9. 

Proposition 6.13. If the series $\sum_{j=1}^{\infty} a_j$ converges absolutely, then it converges. Moreover, the **triangular inequality** $|\sum_{j=1}^{\infty} a_j| \leq \sum_{j=1}^{\infty} |a_j|$ holds.

Proof. Let $\varepsilon > 0$ and pick $N \in \mathbb{N}$, such that for $n \geq m \geq N$,

$$\sum_{j=m}^n |a_j| < \varepsilon.$$

So, by the triangular inequality (exercise 1-42),

$$\left| \sum_{j=m}^n a_j \right| \leq \sum_{j=m}^n |a_j| < \varepsilon.$$

As such, $\sum_{j=1}^{\infty} a_j$ converges. Furthermore, (exercise 2-12)

$$\lim_{n \rightarrow \infty} \left| \sum_{j=1}^n a_j \right| = \left| \sum_{j=1}^{\infty} a_j \right|.$$

It is hence clear that

$$\left| \sum_{j=1}^{\infty} a_j \right| \leq \sum_{j=1}^{\infty} |a_j|.$$



Theorem 6.15 (Comparison Test). Let $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ be series with $0 \leq a_j \leq b_j$ for all $j \in \mathbb{N}$. If $\sum_{j=1}^{\infty} b_j$ converges, then $\sum_{j=1}^{\infty} a_j$ converges, too.

Proof. When $\sum_{j=1}^{\infty} b_j$ converges, the partial sums $\sum_{j=1}^n a_j$ are bounded above by $\sum_{j=1}^{\infty} b_j$. By Lemma 6.3, $\sum_{j=1}^{\infty} a_j$ must converge. 

Theorem 6.16. Let series $\sum_{j=1}^{\infty} a_j$ converges absolutely iff it converges unconditionally.

Proof. Assume $\sum_{j=1}^{\infty} a_j$ converges absolutely. Let σ be a permutation of $\mathbb{N}$ and $\varepsilon > 0$. So, pick $N \in \mathbb{N}$ such that if $n \geq m \geq N$,

$$\sum_{j=m}^n |a_j| < \varepsilon.$$

Fix $M \in \mathbb{N}$, such that for any $i \geq M$ we have $\sigma(i) \geq N$. Further define $u_n := \max\{\sigma^{-1}(j) \mid j \leq n\} \geq n$. For every $n \geq m \geq M$, we see that

$$\sum_{i=m}^n |a_{\sigma(i)}| \leq \sum_{j=N}^{u_n} |a_j| < \varepsilon.$$

As such, $\sum_{i=1}^n a_{\sigma(i)}$ converges. Therefore, $\sum_{j=1}^{\infty} a_j$ converges unconditionally.

Now consider when $\sum_{j=1}^{\infty} a_j$ does not converge absolutely. Let $\{a_{p_k}\}_{k=1}^{\infty}$ and $\{a_{q_k}\}_{k=1}^{\infty}$ be the subsequences of all nonnegative and all negative a_j , respectively.

Wlog,

$$\sum_{k=1}^{\infty} a_{p_k} = \infty \quad \text{and} \quad \sum_{k=1}^{\infty} a_{q_k} = -\infty.$$

Thus, for each n , there is the least natural number ℓ_n such that

$$\sum_{k=1}^{\ell_n} a_{p_k} - \sum_{k=1}^n a_{q_k} \geq n.$$

We hence have the permutation σ defined by

$$\sigma = (\underbrace{p_1, p_2, \dots, p_{\ell_1}}_{1 \text{ to } \ell_1}, \underbrace{q_1, p_{\ell_1+1}, p_{\ell_1+2}, \dots, p_{\ell_2}}_{\ell_1+1 \text{ to } \ell_2}, \underbrace{q_2, \dots, p_{\ell_n+1}, p_{\ell_n+2}, \dots, p_{\ell_{n+1}}}_{\ell_n+1 \text{ to } \ell_{n+1}}, q_{n+1}, \dots).$$

Hence,

$$\sum_{i=1}^{\infty} a_{\sigma(i)} = \infty.$$

does not converge unconditionally



Remark. If exactly one of $\sum_{k=1}^{\infty} a_{p_k}$ or $\sum_{k=1}^{\infty} a_{q_k}$ diverges, then $\sum_{j=1}^{\infty} a_j$ diverges; it does not converge unconditionally.

When that both series converge, $\sum_{j=1}^{\infty} |a_j| \leq \sum_{k=1}^{\infty} a_{p_k} - a_{q_k}$ also converges, a contradiction.

Remark. To be pedantic, we would write $\sigma(1) = p_1$ and

$$\sigma(i+1) = \begin{cases} \sigma(i) + 1 & \text{if } \sigma(i) = p_m \text{ for some } \ell_n \leq m < \ell_{n+1}, \\ p_{\ell_{n+1}} & \text{if } \sigma(i) = q_n \text{ for some } n, \\ q_{n+1} & \text{if } \sigma(i) = p_{\ell_n} \text{ for some } n. \end{cases}$$

Proposition 6.22. Let $\{a_{(i,j)}\}_{i,j=1}^\infty$ be a family of nonnegative numbers. Then, the double series $\sum_{i=1}^\infty \sum_{j=1}^\infty a_{(i,j)}$ converges iff for all bijections $\sigma: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ the sum $\sum_{i=1}^\infty \sum_{j=1}^\infty a_{\sigma(i,j)}$ converges. Furthermore, in this case the values are equal.

Proof. Assume the double series $\sum_{i=1}^\infty \sum_{j=1}^\infty a_{(i,j)} = \sum_{i=1}^\infty |\sum_{j=1}^\infty a_{(i,j)}|$ converges. The preceding theorem guarantees unconditional convergence. So $\sum_{i=1}^\infty \sum_{j=1}^\infty a_{\sigma(i,j)}$ converges to the same value. (The necessary permutation on $\mathbb{N}$ is obvious.)

The converse is trivial. Simply let σ be the identity function on $\mathbb{N} \times \mathbb{N}$. 

Exercise 6.19. Give an example of an absolutely convergent series $\sum_{j=1}^\infty a_j$ so that $\sum_{j=1}^\infty a_j \neq \sum_{j=1}^\infty |a_j|$.

Proof. Consider the geometric sequence $\{(-1/2)^j\}_{j=1}^\infty$. Then, the sum

$$\sum_{j=1}^\infty \left(-\frac{1}{2}\right)^j = -\frac{1}{2}$$

whilst

$$\sum_{j=1}^\infty \left|-\frac{1}{2}\right|^j = \frac{1}{2}.$$



Exercise 6.21. Determine which of the following series converges. If it converges, determine if it converges absolutely.

$$(a) \sum_{j=1}^\infty \frac{1}{j!}, \quad (b) \sum_{j=1}^\infty \frac{(-1)^j}{j+\sqrt{j}}, \quad (c) \sum_{j=1}^\infty \frac{(-1)^j}{4^j}$$

Proof.

(a) Notice that $0 \leq 1/j! \leq (1/2)^{j-1}$ for all j . Hence, by the comparison test, $\sum_{j=1}^\infty \frac{1}{j!}$ converges (absolutely).

(b) For each j , we have

$$0 \leq \frac{1}{2j} \leq \frac{1}{j+\sqrt{j}}.$$

By the comparison test, since the harmonic series diverges, so must $\sum_{j=1}^\infty \frac{(-1)^j}{j+\sqrt{j}}$. Hence, absolute convergence is impossible. But regular convergence is possible.

Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$, such that $\varepsilon > N^{-1}$. So, if $n \geq m \geq N$,

$$\begin{aligned} \left| \sum_{j=m}^n \frac{(-1)^j}{j + \sqrt{j}} \right| &= \left| \sum_{j=0}^{n-m} \frac{(-1)^{m+j}}{m+j + \sqrt{m+j}} \right| \\ &\leq \sum_{\substack{j=0 \\ j \text{ even}}}^{n-m} \frac{1}{m+j + \sqrt{m+j}} - \sum_{\substack{j=0 \\ j \text{ odd}}}^{n-m} \frac{1}{m+j+1 + \sqrt{m+j+1}} \\ &\leq \frac{1}{m + \sqrt{m}} < \varepsilon. \end{aligned}$$

(c) We see that this is just a geometric series with constant ratio $-1/4$. Thus, it converges (absolutely).



Exercise 6.22 (Limit Comparison Test for series). Let $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ be series with positive terms. Prove that if $\lim_{j \rightarrow \infty} a_j/b_j = c > 0$, then either both series converge or both series diverge.

Proof. Since $c > 0$, we can assume (for the sake of argument) wlog that $\sum_{j=1}^{\infty} b_j$ converges. Moreover, notice $\{a_j/b_j\}_{j=1}^{\infty}$ is bounded from above by some $u > 0$. As such, $0 < a_j \leq ub_j$ for all j . Therefore, $\sum_{j=1}^{\infty} a_j$ converges.



Note. The importance of $c > 0$ should be noted. If $c = 0$, then $\lim_{j \rightarrow \infty} b_j/a_j = \infty$. As such, the “wlog” part of the proof fails to hold. It is possible that $\sum_{j=1}^{\infty} a_j$ converges, while $\sum_{j=1}^{\infty} b_j$ diverges.

Example. Consider $a_j = 2^{-j}$ and $b_j = 2^j$. Then, $\lim_{j \rightarrow \infty} a_j/b_j = \lim_{j \rightarrow \infty} 2^{-2j} = 0$. But it is clear that $\sum_{j=1}^{\infty} a_j = 1$, whilst $\sum_{j=1}^{\infty} b_j = \infty$.

The definition of the permutation σ in our proof of theorem 6.16 was a little inelegant. Let's try defining a little tool to help us have better notation (hopefully).

Definition. Given a sequence of permutations $\sigma_i := (a_{i1}, a_{i2}, \dots, a_{ik_i})$ of k_i numbers, we define

$$(\sigma_1, \sigma_2, \dots) := (\underbrace{a_{11}, a_{12}, \dots, a_{1k_1}}_{\sigma_1}, \underbrace{a_{21}, a_{22}, \dots, a_{2k_2}}_{\sigma_2}, \dots).$$

Exercise 6.23. Let $\sum_{j=1}^{\infty} a_j$ be a conditionally convergent series and let $z \in \mathbb{R}$. Prove that there is a bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ such that $\sum_{j=1}^{\infty} a_{\sigma(j)} = z$.

Proof. As in theorem 6.16, we let $\{a_{p_k}\}_{k=1}^{\infty}$ and $\{a_{q_k}\}_{k=1}^{\infty}$ be the subsequences of all nonnegative and all negative a_j , respectively. Conditional convergence implies that $\sum_{j=1}^{\infty} a_{p_k} = \infty$, and $\sum_{j=1}^{\infty} a_{q_k} = -\infty$. Hence, we define ν_n recursively by

$\nu_0 := 0$, and having ν_n be the least natural number with

$$\sum_{k=1}^{\nu_n} a_{p_k} + \sum_{k=1}^{\mu_{n-1}} a_{q_k} > z,$$

where μ_{n-1} is the least natural number with

$$\sum_{k=1}^{\nu_{n-1}} a_{p_k} + \sum_{k=1}^{\mu_{n-1}} a_{q_k} < z.$$

Now define $P_n := (p_{\nu_n+1}, p_{\nu_n+2}, \dots, p_{\nu_{n+1}})$ and $Q_n := (q_{\mu_n+1}, q_{\mu_n+2}, \dots, q_{\mu_{n+1}})$.

We obtain the permutation

$$\sigma := (P_0, Q_0, P_1, Q_1, \dots, P_n, Q_n, \dots).$$

Let $\varepsilon > 0$ and note that $\lim_{k \rightarrow \infty} a_{p_k} = \lim_{k \rightarrow \infty} a_{q_k} = 0$. Thus, there is some $K \in \mathbb{N}$ such that if $k \geq K$, we have

$$0 \leq a_{p_k} < \varepsilon \quad \text{and} \quad 0 < -a_{q_k} < \varepsilon.$$

Consequently,

$$\left| \sum_{j=1}^k a_{\sigma(j)} - z \right| < \varepsilon.$$



Exercise 6.24. Let $\{a_{(i,j)}\}_{i,j=1}^{\infty}$ be a family of real numbers so that the double series $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{(i,j)}|$ converges. Prove that $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}$ converges to a number L and for all bijections $\sigma: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ the sum $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{\sigma(i,j)}$ converges to the same number L .

Proof. Recall that $0 \leq |\sum_{j=1}^{\infty} a_{(i,j)}| \leq \sum_{j=1}^{\infty} |a_{(i,j)}|$ always holds. So, the series

$$\sum_{i=1}^{\infty} \left| \sum_{j=1}^{\infty} a_{(i,j)} \right| \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{(i,j)}|$$

converges absolutely, and hence, unconditionally to some limit L .



Some Set Theory

7.1 The Algebra of Sets

Exercise 7.5. Let X be a set. An **algebra** is a set of sets $\mathcal{A} \subseteq \mathcal{P}(X)$ such that $\emptyset \in \mathcal{A}$, if $A \in \mathcal{A}$, then $X \setminus A \in \mathcal{A}$, and if $A_j \in \mathcal{A}$ for all $j = 1, \dots, n$, then $\bigcup_{j=1}^n A_j \in \mathcal{A}$.

- (a) Prove that if $A_j \in \mathcal{A}$ for all $j = 1, \dots, n$, then $\bigcap_{j=1}^n A_j \in \mathcal{A}$.
- (b) Let X be a set. Prove that the power set of X is an algebra.
- (c) Let X be a set. Prove that $\mathcal{A} := \{A \subseteq X \mid A \text{ or } X \setminus A \text{ is finite}\}$ is an algebra.
- (d) Prove that an algebra need not contain countable unions of its elements.

Proof.

- (a) Notice that $X - \bigcap_{j=1}^n A_j = \bigcup_{j=1}^n (X - A_j) \in \mathcal{A}$. So, $\bigcap_{j=1}^n A_j \in \mathcal{A}$ follows.
- (b) We see that $\emptyset \subseteq X$, and $X - A \subseteq X$ for all $A \subseteq X$. Moreover, a union of subsets of X is still a subset of X . Hence, the power set of X is indeed an algebra.
- (c) Since $\emptyset \subseteq X$ is obviously finite, $\emptyset \in \mathcal{A}$. For $A \in \mathcal{A}$, clearly A or $X - A$ is finite. Thus, $X - A \in \mathcal{A}$. Now let $A_1, A_2, \dots, A_n \in \mathcal{A}$. If A_j is infinite for some j , then $X - \bigcup_{j=1}^n A_j \subseteq X - A_j$ is finite. When each A_j is finite, $\bigcup_{j=1}^n A_j$ is finite. Either ways, $\bigcup_{j=1}^n A_j \in \mathcal{A}$ holds.
- (d) Consider the set $\mathcal{F}$ of all finite subsets of $\mathbb{N}$. It is easily verified to be an algebra. However, the countable union $\bigcup_{j=1}^{\infty} \{j\} = \mathbb{N} \notin \mathcal{F}$.



7.2 Countable Sets

Lemma. Let $\{a_{(i,j)}\}_{j=1}^{\infty}$ be a family of numbers, such that the double series $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}$ converges.

Exercise 7.17. Let $\{a_{(i,j)}\}_{j=1}^{\infty}$ be a family of nonnegative numbers. Then the double series $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}$ converges iff for all bijections $\sigma: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ the sum $\sum_{i=1}^{\infty} a_{\sigma(i)}$ converges. Furthermore, in this case the values are equal.

Proof. Assume $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}$ and let $\sigma: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ be a bijection. Fix n and m . So, define $\lceil := \max\{(\pi_1 \circ \sigma)(\iota) \mid \iota \leq n\}$ and

$$\mathfrak{I}_i := \max\{(\pi_2 \circ \sigma)(j) \mid (\pi_1 \circ \sigma)(\iota) = i \text{ \& } \iota \leq n\}.$$

Hence,

$$0 \leq \sum_{i=1}^n a_{\sigma(i)} \leq \sum_{i=1}^{\lceil} \sum_{j=1}^{\mathfrak{I}_i} a_{(i,j)} \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}.$$

The series $\sum_{i=1}^{\infty} a_{\sigma(i)}$ must converge.

Conversely, suppose $\sum_{i=1}^{\infty} a_{\sigma(i)}$ converges for all bijections $\sigma: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ and fix $n \in \mathbb{N}$. Now letting $\mathcal{M}_i := \max\{\sigma^{-1}(i, j) \mid j \leq m\}$, we have that

$$0 \leq \sum_{j=1}^m a_{(i,j)} \leq \sum_{i=1}^{\mathcal{M}_i} a_{\sigma(i)} \leq \sum_{i=1}^{\infty} a_{\sigma(i)}.$$

Therefore, $\sum_{j=1}^{\infty} a_{(i,j)}$ converges for all i .

Now let $\varepsilon > 0$. Pick $M \in \mathbb{N}$ such that, for every $1 \leq m \leq n$,

$$\sum_{j=1}^{\infty} a_{(i,j)} - \sum_{j=1}^m a_{(i,j)} < \frac{\varepsilon}{n}.$$

Thus, for $\mathcal{M} := \max\{\sigma^{-1}(i, j) \mid i \leq n \text{ \& } j \leq m\}$, we see that

$$0 \leq \sum_{i=1}^n \sum_{j=1}^{\infty} a_{(i,j)} < \varepsilon + \sum_{i=1}^n \sum_{j=1}^m a_{(i,j)} \leq \varepsilon + \sum_{i=1}^{\mathcal{M}} a_{\sigma(i)} \leq \varepsilon + \sum_{i=1}^{\infty} a_{\sigma(i)}.$$

As such, the series $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}$ converges.

In fact, since

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)} \leq \sum_{i=1}^{\infty} a_{\sigma(i)},$$

it holds that

$$\sum_{i=1}^{\infty} a_{\sigma(i)} = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{(i,j)}.$$



7.3 Uncountable Sets

Exercise 7.23. Let $F: [a, b] \rightarrow \mathbb{R}$ be a *nondecreasing* bounded function. Prove that F can have at most countably many discontinuities.

Hint. Suppose the set of discontinuities is uncountable and use exercise 3.21 to conclude that there must be an ε so that the set of all x with $\lim_{z \rightarrow x^+} F(z) - \lim_{z \rightarrow x^-} F(z) > \varepsilon$ is uncountable.

Proof. Let $x \in [a, b]$, $L := \sup f[a, x]$ and $\varepsilon > 0$. Then, there is $y \in [a, x]$ for which $L - f(y) < \varepsilon$. Consequently, since F is nondecreasing, for any $x - z < x - y$ we have $L - f(z) \leq L - f(y) < \varepsilon$. The left limit $\lim_{z \rightarrow x^-} f(z) = L$ exists, and similarly, so does the right limit $\lim_{z \rightarrow x^+} f(z)$. This implies only jump discontinuities are present.

(Work in progress)



Proof (Hint used). Let $A_\varepsilon := \{x \in [a, b] \mid F(x^+) - F(x^-) > \varepsilon\}$. Notice that $A_{1/n}$ being finite for all n implies the countability of D_f — which comprises of solely jump discontinuities. So, if D_f were uncountable, A_ε must be infinite for some $\varepsilon > 0$. But F is bounded, a contradiction.



Exercise 7.24. Prove that for every countable subset $A \subseteq \mathbb{R}$ there is a **nondecreasing** function $f: \mathbb{R} \rightarrow [0, 1]$ that is continuous on $\mathbb{R} \setminus A$ and discontinuous at every $a \in A$.

Hint. With $A = \{a_n \mid n \in \mathbb{N}\}$ set $f(x) := \sum_{a_n < x} 2^{-n}$.

Proof. Failed attempt.

Wlog, consider A being countably infinite. Let there g be a bijection from $\mathbb{N}$ to A . Define the injection $h: \mathbb{N} \rightarrow [0, 1]$ by $h(0) := 1/2$ and the following.

- i. If $g(k+1) < g(i)$ for all $i \leq k$, then let $h(n+1) := h(\ell)/2$, where $h(\ell) := \min\{h(i) \mid i \leq k\}$.
- ii. When $g(k+1) > g(i)$ for every $i \leq k$, let $h(n+1) := \frac{h(M)+1}{2}$, where $h(M) := \max\{h(i) \mid i \leq k\}$.
- iii. Otherwise, let $h(k+1) = \frac{h(n)+h(m)}{2}$ for the largest $g(n) < g(k+1)$ and least $g(m) < g(k+1)$ such that $n, m \leq k$.

Now define the function $f: \mathbb{R} \rightarrow [0, 1]$ by $f(a) = (h \circ g^{-1})(a)$ for $a \in A$, and for $x \in \mathbb{R} - A$,

$$f(x) = \begin{cases} \inf\{f(a) \mid a > x\} & \text{if } a > x \text{ for some } a \in A, \\ 1 & \text{otherwise.} \end{cases}$$

We omit the tedious proof that f is nondecreasing for brevity's sake.

To show continuity on $\mathbb{R} - A$, let $x \in \mathbb{R} - A$ and $\varepsilon > 0$. Wlog, $f(x) - f(a) < \varepsilon$ for some $a > x$. Since f is nondecreasing, for all $z - x < a$ we have $f(z) - f(x) < \varepsilon$.

Thus, $\lim_{z \rightarrow x^+} f(z) = f(x)$. If $A \cap (x - \delta, x) = \emptyset$ for some $\delta > 0$, then for any $z \in (x - \delta, x)$ we have $f(z) = f(x)$. Otherwise, for each $\delta > 0$, there is some $b \in A \cap (x - \delta, x)$. Let $\varphi(1)$ be the least natural number, such that there exists $i, j \leq \varphi(1)$, with $f(i) < f(x)$ and $f(j) > f(x)$. Also let

$$\varphi(k+1) := h^{-1} \left(\frac{l+u}{2} \right),$$

where $l := \max\{(f \circ h)(i) < f(x) \mid i \leq k\}$ and $u := \min\{(f \circ h)(i) > f(x) \mid i \leq k\}$. Suppose, for contradiction, there is some least $\alpha \in \mathbb{N}$, such that

$$\varphi[\mathbb{N}] \cap \left(f(x) - \frac{1}{\alpha}, f(x) \right) = \emptyset.$$

Then, for some k ,

$$(f \circ h)(k) \in \left(f(x) - \frac{1}{\alpha}, f(x) - \frac{1}{\alpha-1} \right].$$

We can also pick $(f \circ h)(k') \in (f(x), 2f(x) - f(k))$. Therefore,

$$\frac{f(a) + f(k)}{2} \in \left(x - \frac{1}{\alpha}, x \right),$$

As such, for $\mathcal{M} := \max(k, k')$, we have

$$(f \circ \varphi)(\mathcal{M}) \in \left(x - \frac{1}{\alpha}, x \right).$$

A contradiction. Whence, it is clear that the right limit exists. In fact, $\lim_{z \rightarrow x^-} f(z) = f(x)$.

Showing f is discontinuous at every $a \in A$ is straightforward.

It was in fact, not straightforward.

(Work in progress)



Proof (Hint used). Choose an enumeration $\{a_n\}$ of A and $f(x) := \sum_{a_n < x} 2^{-n}$. Let $\{a_{m_k}\}$ and $\{a_{n_k}\}$ be the subsequences with $m_k < x < n_k$. For $\varepsilon > 0$ and $x \in \mathbb{R} - A$, pick $K \in \mathbb{N}$ such that

$$\sum_{k > K} 2^{-m_k} < \varepsilon \quad \text{and} \quad \sum_{k > K} 2^{-n_k} < \varepsilon.$$

Now, the continuity of f at x follows from selecting $\delta := \min\{|x - a_{m_k}|, |x - a_{n_k}|\}_1^K$. As for $a = a_n$, notice $f(a^+) - f(a^-) = 2^{-n}$. So, $D_f = A$ as required.

✓checked by Bezier



Exercise 7.25. Cantor sets.

(a) Prove that for any sequence $Q = \{q_n\}_{n=1}^\infty$ of numbers $q_n \in (0, 1/2)$ there is a

- bijjective function from C^Q to the set of all sequences of zeroes and ones.
- (b) Prove that C^Q is uncountable.
 - (c) Prove that the set of endpoints of the intervals $I_{j,n}^Q$ that make up the C_n^Q is countable.
 - (d) Prove that every $x \in C^Q$ is the limit of a sequence of endpoints of the intervals $I_{j,n}^Q$.

Proof.

- (a) Let $a = \{a_n\}$ be a sequence of zeros and ones. Pick a sequence $\{x_n\}$ with

$$x_1 \in [0, 1] \quad \text{and} \quad x_{n+1} \in \begin{cases} L_{q_{n+1}}[I_{i_n,n}^Q] & \text{if } a_{n+1} = 0, \\ R_{q_{n+1}}[I_{i_n,n}^Q] & \text{if } a_{n+1} = 1, \end{cases}$$

where $x_n \in I_{i_n,n}^Q$. Clearly, $\{x_n\}$ is Cauchy and converges to some (unique) x_a , since $x_n, x_{n+1}, \dots \in I_{i_n,n}$ and $|I_{i_n,n}| \rightarrow 0$. The map $a \rightarrow x_a$ must be injective: $|x_a - x_b| \geq (1 - 2q_n)|I_n| > 0$ if $a_n \neq b_n$ for some n . Surjectivity is also apparent from $C^Q := \bigcap_n C_n^Q = \bigcap_n \bigcup_{i=1}^{2^n} I_{i,n}^Q$.

- (b) It follows from (a) that $|C^Q| = 2^{\aleph_0} > \aleph_0$.
- (c) Clearly, this set of endpoints has cardinality 2^{n+1} , as C_n^Q is the union of 2^n disjoint intervals.
- (d) Similar to (a).



$|\cdot|$ is used to represent both the cardinality of a set and the outer Lebesgue measure of an interval. From context, it should be clear which is in use.

The Riemann Integral II

8.1 Outer Lebesgue Measure

Proposition 8.3. Countable subsets of $\mathbb{R}$ have outer Lebesgue measure 0.

Proof. Wlog, let $C \subseteq \mathbb{R}$ be a countably infinite set. Then, there is a bijection $f: \mathbb{N} \rightarrow C$. So, for each $\varepsilon > 0$ we define $h_\varepsilon: \mathbb{N} \rightarrow \mathcal{P}(\mathbb{R})$ by

$$h_\varepsilon(j) := (f(j) - \varepsilon^j, f(j) + \varepsilon^j).$$

We see that for $\varepsilon \in (0, 1)$:

$$\sum_{j=1}^{\infty} |h_\varepsilon(j)| = \sum_{j=1}^{\infty} 2\varepsilon^j = \frac{2\varepsilon}{1-\varepsilon} = -2 + \frac{2}{1-\varepsilon}.$$

Furthermore,

$$\lim_{\varepsilon \rightarrow 0} -2 + \frac{2}{1-\varepsilon} = 0.$$

Therefore, $\lambda(C) = 0$.



Proposition 8.5. Let $a, b \in \mathbb{R}$ and $a < b$. Then, $\lambda([a, b]) = b - a$.

Proof. For each $\varepsilon > 0$, let $I_1 := (a - \varepsilon, b + \varepsilon)$, and for $j \geq 2$, $I_j := (0, \varepsilon^j)$. Therefore, given $\varepsilon \in (0, 1)$,

$$\sum_{j=1}^{\infty} |I_j| = b - a + 2\varepsilon + \frac{\varepsilon}{1-\varepsilon}.$$

As $\varepsilon \rightarrow 0$, this converges to $b - a$. i.e. $\lambda([b - a]) \leq b - a$.

Now let $I_j = (a_j, b_j)$ be a sequence of intervals that covers $[a, b]$. So by the Heine-Borel Theorem, $[a, b] \subseteq \bigcup_{j=1}^n I_j$ for some n . Wlog, $a_j \leq a_{j+1} \leq b_j \leq b_{j+1}$ for each

j . Therefore,

$$\sum_{j=1}^n |I_j| \geq b_1 - a_1 + \sum_{j=2}^n b_j - b_{j-1} \geq b_n - a_1 \geq b - a.$$

i.e. $\lambda([b - a]) \geq b - a$.



Theorem 8.6 (The *properties of the outer Lebesgue measure* λ). With ∞ defined to be greater than all real numbers and the sum of a divergent series of nonnegative numbers being ∞ we have the following.

- (a) $\lambda(\emptyset) = 0$.
- (b) If $A \subseteq B$, then $\lambda(A) \leq \lambda(B)$.
- (c) Outer Lebesgue measure is *countably subadditive*. That is, for all sequences $\{A_n\}_{n=1}^{\infty}$ of subsets $A_n \subseteq \mathbb{R}$ the inequality

$$\lambda\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \lambda(A_n)$$

holds.

Proof.

- (a) Let $\varepsilon > 0$ and $I_j = (0, \varepsilon 2^{-j})$. Then, $\sum_{j=1}^{\infty} I_j \leq \varepsilon$. Thus, $\lambda(\emptyset) = 0$.
- (b) Let $\varepsilon > 0$, and $\{I_j\}_{j=1}^{\infty}$ be a sequence of open intervals that covers B , such that $\sum_{j=1}^{\infty} |I_j| < \lambda(B) + \varepsilon$. Hence, it covers A , which implies

$$\lambda(A) \leq \sum_{j=1}^{\infty} |I_j| < \lambda(B) + \varepsilon.$$

As such, $\lambda(A) \leq \lambda(B)$.

- (c) Let $\varepsilon > 0$ and $\sigma: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ be a bijection. For each n , choose a sequence of open intervals $\{I_{(n,j)}\}_{j=1}^{\infty}$ that cover A_n , such that $\sum_{j=1}^{\infty} |I_{(n,j)}| < \lambda(A_n) + \varepsilon 2^{-n}$. Notice that $\{I_{\sigma(i)}\}_{i=1}^{\infty}$ covers $\bigcup_{n=1}^{\infty} A_n$. By exercise 7.17,

$$\lambda\left(\sum_{n=1}^{\infty} A_n\right) \leq \sum_{i=1}^{\infty} |I_{\sigma(i)}| = \sum_{n=1}^{\infty} \sum_{j=1}^{\infty} |I_{(n,j)}| < \varepsilon + \sum_{n=1}^{\infty} \lambda(A_n).$$

Consequently,

$$\lambda\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \lambda(A_n).$$



Question. Is the outer Lebesgue measure (countably) additive for disjoint sets?

Claim. We have $\lambda(A - B) = \lambda(A) - \lambda(B)$ iff A contains B .

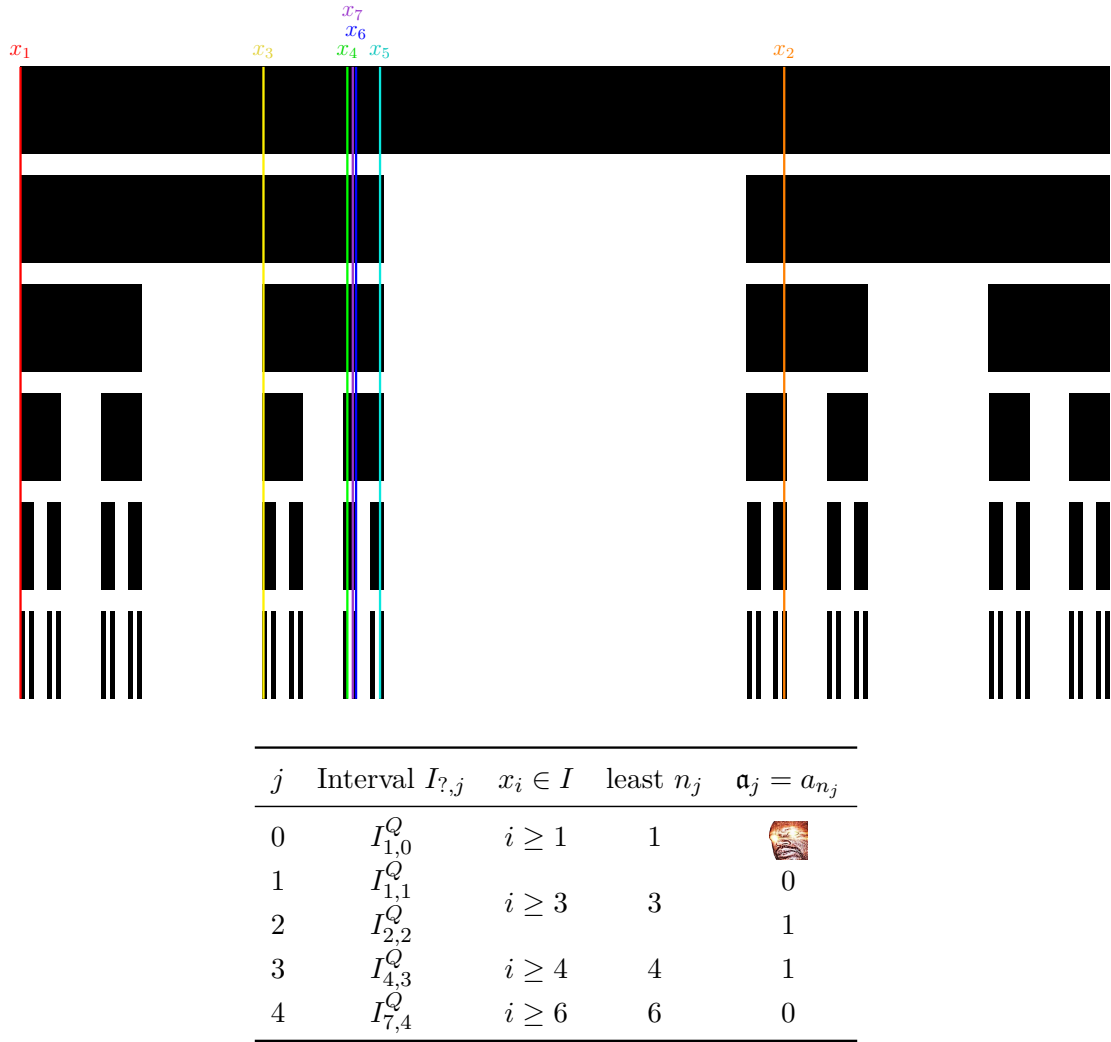


Figure 8.1: An illustration of the procedure in 8.3(c) to prove that all Cantor sets are closed.

Exercise 8.3. Let $Q = \{q_n\}_{n=1}^\infty$ be a sequence of numbers $q_n \in (0, 1/2)$ and let C^Q be the associated *Cantor set* as in Definition 7.22. We will use the notation of Definition 7.22 throughout this exercise.

- Prove that $\lambda(C_n^Q) = \prod_{j=1}^n 2q_j$.
- Prove that $\{\prod_{j=1}^n 2q_j\}_{n=1}^\infty$ converges.
- Prove that $\lambda(C^Q) = \lim_{n \rightarrow \infty} \prod_{j=1}^n 2q_j$.

Hint. For “ $\geq$ ”, first consider a family $\mathcal{I}$ of open intervals so that $C^Q \subseteq \bigcup \mathcal{I}$. Prove that

$$C := \left\{ x \in [0, 1] \mid C^Q \cap [0, x] \not\subseteq \bigcup_{j=1}^n I_j \text{ for all } I_1, I_2, \dots, I_n \in \mathcal{I} \right\} = \emptyset,$$

by assuming it is not empty and showing that $\inf C \in C^Q$, which leads to a contradiction similar to the proof of the Heine-Borel Theorem. (It also helps to look ahead to the proof of Lemma 8.11 for a similar argument.) Then

show that for any countable family $\{I_j\}_{j=1}^\infty$ with $C^Q \subseteq \bigcup I_j$ there are intervals $I_{j_1}, \dots, I_{j_m}$ with $C^Q \subseteq \bigcup_{k=1}^m I_{j_k}$. Conclude that there must be an $n \in \mathbb{N}$ so that $C_n^Q \subseteq \bigcup_{k=1}^m I_{j_k}$.

- (d) Prove that for any $q \in (0, 1/2)$ the constant sequence $Q = \{q\}_{n=1}^\infty$ yields a Cantor set C^Q of measure zero.

Note. By exercise 7-25(b) in section 7.3 Cantor sets are uncountable. This means there are uncountable sets of measure zero.

- (e) Use $Q = \left\{ \frac{2^{n+1}-1}{2^{n+2}} \right\}_{n=1}^\infty$ to prove that there are Cantor sets that are not of measure zero.
- (f) Prove that there are Cantor sets whose Lebesgue measure is arbitrarily close to 1.

Proof.

- (a) If $n = 0$, then $\lambda([0, 1]) = \lambda(C_n^Q) = \prod_{j=1}^0 2q_j = 1$. So, assume $n = k$ and consider $n = k + 1$. It is clear that, $\lambda(I_{i,m}^Q) = \lambda(I_{j,m}^Q)$ for all m and $i, j \leq k$. Therefore, $\lambda(I_{i,k}^Q) = \prod_{j=1}^k q_j$. Thus, $\lambda(I_{i,k+1}^Q) = \prod_{j=1}^{k+1} q_j$, and hence, $\lambda(C_{k+1}^Q) = \prod_{j=1}^{k+1} 2q_j$.
- (b) Since $2q_j < 1$ for each j , this sequence is decreasing. Furthermore, it is bounded below by 0, and hence, must converge.
- (c) Let $\{x_i\}$ be a sequence in C^Q . Recall, from 7.25(a), that each x_i corresponds (uniquely) to some $\{a_{i,j}\}_{j=1}^\infty \in \mathbb{N}\{0, 1\}$. Assuming $\{x_i\}$ converges, $\mathbf{a}_j := \lim_i a_{i,j}$ exists. In fact, $\{a_{i,j}\}_{i=n_j}^\infty$ is constant for large n_j (Figure 8.1). Therefore, $x_i \rightarrow \mathbf{x}_\mathbf{a} \in C^Q$ is clear: $|x_i - \mathbf{x}_\mathbf{a}| < 1/2^J$, for $i \geq \max_{1 \leq j \leq J} n_j$. Since C^Q is closed and bounded, it is compact.
- Now let $\{I_n = (b_n, c_n)\}$ cover C^Q and $\{I_{n_k}\}_1^K$ be a finite subcover. Pick $n \in \mathbb{N}$ with

$$1/2^n < \inf\{|x - b_{n_k}|, |x - c_{n_k}| : x \in C^Q \text{ and } 1 \leq k \leq K\}.$$

So $I_{i,n}^Q \subseteq I_{n_k}$ for $I_{i,n}^Q \cap I_{n_k} \neq \emptyset$. Consequently,

$$C^Q \subseteq C_n^Q \subseteq \bigcup I_n$$

yields $\lambda(C^Q) = \lim_{n \rightarrow \infty} \lambda(C_n^Q) = \lim_{n \rightarrow \infty} \prod_{j=1}^n 2q_j$.

- (d) Since $2q < 1$,

$$\lambda(C^Q) = \lim_{n \rightarrow \infty} (2q)^n = 0$$

follows from (c).

- (e) Suppose that $\prod_{n=1}^m \frac{2^{n+1}-1}{2^{n+1}} > 1/2 + 1/2^m$. Observe that

$$\prod_{n=1}^{m+1} \frac{2^{n+1}-1}{2^{n+1}} > \left(1 - \frac{1}{2^{m+2}}\right) \left(\frac{1}{2} + \frac{1}{2^m}\right) \geq \frac{1}{2} + \frac{1}{2^{m+1}};$$

the last inequality follows from $3/8 > 1/4 \geq \frac{1}{2^{m+2}}$. So, $\lambda(C^Q) \geq 1/2$.

(f) Now select $Q_\varepsilon = \left\{ \frac{2^{n+1}-\varepsilon}{2^{n+2}} \right\}$ and assume $\prod_{n=1}^m \frac{2^{n+1}-\varepsilon}{2^{n+1}} \geq 1 - \varepsilon + 1/2^m$. From $\varepsilon^2 - \varepsilon(1 + 1/2^m) + 2 \geq 0$, we recover

$$\prod_{n=1}^{m+1} \frac{2^{n+1}-\varepsilon}{2^{n+1}} \geq \left(1 - \frac{\varepsilon}{2^{m+2}}\right) \left(1 - \varepsilon + \frac{1}{2^m}\right) \geq 1 - \varepsilon + \frac{1}{2^{m+1}}.$$

Hence, $C^{Q_\varepsilon} \geq 1 - \varepsilon$.



In hindsight, I should have thought about compactness for 8.3(c), given that open interval covers of a set define the outer Lebesgue measure. But, somehow, that eluded my mind.

Question. For each $\ell \in (0, 1)$, is there a Cantor set C^Q with outer Lebesgue measure ℓ ?

Theorem (Hausdorff-Alexandroff). Every compact metric space is a continuous image of any Cantor set C^Q .

Question. Is there a function $f: \mathbb{R} \rightarrow \mathbb{R}$ whose set of infinite discontinuities $\tilde{\mathcal{D}}_f^\infty$ is of positive measure?

Proof. Define $f: [0, 1] \rightarrow \mathbb{R}$ by

$$f(x) := \begin{cases} 0 & \text{if } x \notin \mathbb{Q}, \\ q & \text{if } p \text{ and } q \text{ are coprime such that } x = p/q. \end{cases}$$

Pick an irrational number x and a sequence of rational numbers $p_n/q_n \rightarrow x$; wlog, p_n and q_n are coprime. We must have $q_n \rightarrow \infty$, lest $\{p_n/q_n\}$ contains only finitely many unique rationals; $p_n/q_n \not\rightarrow x$. Consequently, $\tilde{\mathcal{D}}_f^\infty = [0, 1] - \mathbb{Q}$, which has positive measure 1.



Exercise 8.5. Prove that if $f: [a, b] \rightarrow \mathbb{R}$ is continuous and $\lambda(\{x \in [a, b] : f(x) \neq 0\}) = 0$, then $f(x) = 0$ for all $x \in [a, b]$.

Proof. Suppose that f is continuous and attains a zero at some $c \in [a, b]$; let $l := \sup\{x < c \mid f(x) = 0\}$ and $u := \inf\{x > c \mid f(x) = 0\}$. By continuity, $l < c < u$. Hence, $\lambda(\{x \in [a, b] : f(x) \neq 0\}) \geq \lambda(l, u) = u - l > 0$.



Exercise 8.6. Prove that if $f, g: [a, b] \rightarrow \mathbb{R}$ are continuous almost everywhere, then $f + g$ is continuous almost everywhere.

Proof. If f and g are both continuous at a point, then so is $f + g$ at the same point. Hence, the set of discontinuities of $f + g$ is a subset of the union of sets of discontinuities of f and g .



Claim. Any uncountable set A that is dense in $[0, 1]$ have nonzero outer Lebesgue measure.

8.2 Lebesgue's Criterion for Riemann Integrability

Theorem 8.8. The bounded function $f: [a, b] \rightarrow \mathbb{R}$ is continuous at $x \in [a, b]$ iff $\omega_f(x) = 0$.

Proof. Assume f is continuous at x ; let $\varepsilon > 0$ and pick $\delta > 0$, such that $|f(x) - f(y)| < \varepsilon$ if $|x - y| < \delta$. Then, $\omega_f(x) \leq \omega_f(x - \delta, x + \delta) \leq \varepsilon$. Hence, $\omega_f(x) = 0$. Conversely, consider when $\omega_f(x) = 0$. Then, there is an open interval (a, b) containing x , with $|f(x) - f(y)| \leq \omega_f(a, b) < \varepsilon$ for all $y \in (a, b)$. Continuity follows trivially. 

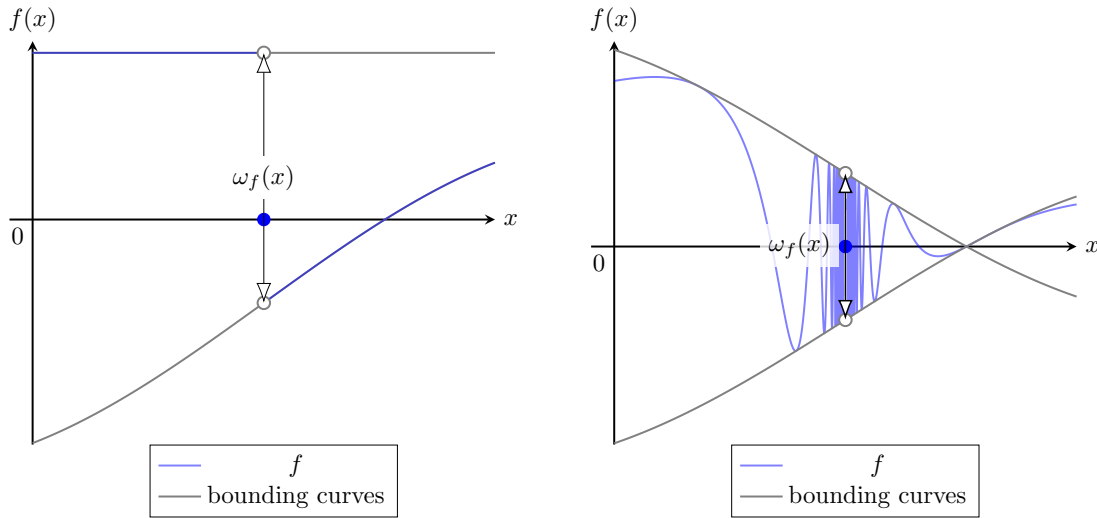
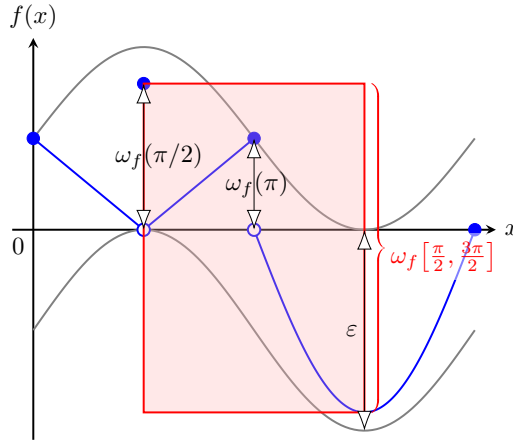


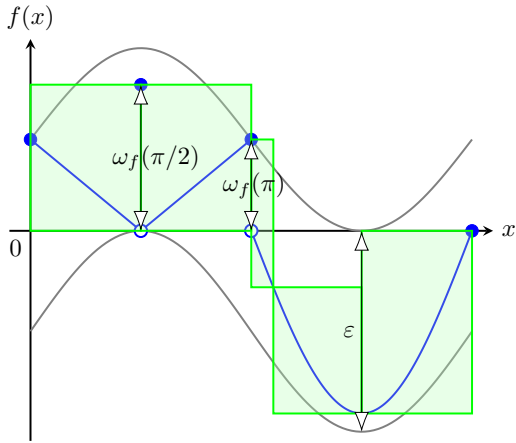
Figure 8.2: An illustration of theorem 8.8.

Lemma 8.9. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. If $\omega_f(x) < \varepsilon$ for all $x \in [a, b]$, then there is a partition P of $[a, b]$ so that $U(f, P) - L(f, P) < \varepsilon|b - a|$.

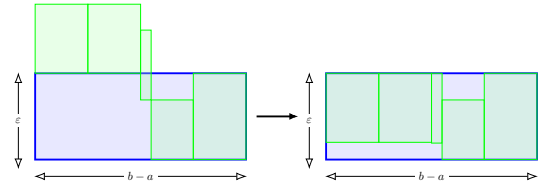
Proof. Suppose that $\omega_f(x) < \varepsilon$ for all $x \in [a, b]$. For each x , pick an open interval (a_x, b_x) over which the oscillation of f is less than ε . By compactness, there is a finite subcover $\{(a_{x_i}, b_{x_i})\}_0^n$ of $[a, b]$, where $x_0 < \dots < x_n$ wlog. Since $M_i - m_i < \varepsilon$, choosing $P = \{x_i\}_0^n$ completes the proof. 



(a) An **interval** over which the oscillation of f exceeds ε .



(b) A **partition** $\{x_i\}_0^5$ satisfying $\omega_f(x_{i-1}, x_i) < \varepsilon$.



(c) A visual representation of how $U(f, P) - L(f, P) < \varepsilon|b - a|$.

Figure 8.3: An illustration of lemma 8.9.

Lemma 8.10. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. If $\lambda(\{x \in [a, b] : \omega_f(x) > 0\}) > 0$, then f is not Riemann integrable.

Proof. Consider when $\lambda(\mathcal{D}_f) > 0$, where $\mathcal{D}_f$ is the set of discontinuities of f . Notice there is $\varepsilon > 0$ with $\lambda(B_\varepsilon) > 0$, lest $\lambda(\mathcal{D}_f) \leq \sum B_{1/n} = 0$. Let $P = \{x_i\}_0^N$ be a partition and pick $\{x_{i_j}\}_0^n$, such that $B_\varepsilon \cap [x_{i-1}, x_i] \neq \emptyset$ iff $i = i_j$ for some j . Since $M_{i_j} - m_{i_j} \geq \varepsilon$,

$$U(f, P) - L(f, P) \geq \sum_{j=1}^n (M_{i_j} - m_{i_j}) \Delta x_{i_j} \geq \varepsilon \lambda(B_\varepsilon) > 0.$$

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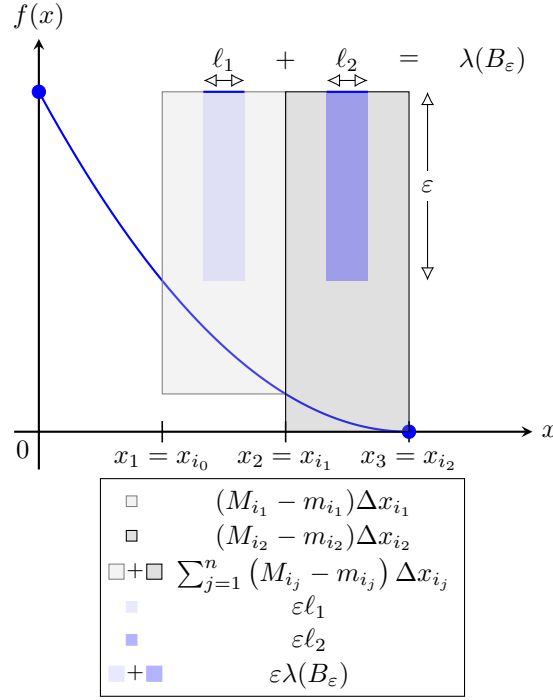


Figure 8.4: An illustration of lemma 8.10. Notice that each blue rectangle is always contained within the corresponding gray rectangle. As such,

$$0 < \varepsilon \lambda(B_\varepsilon) \leq \sum_{j=1}^n (M_{i_j} - m_{i_j}) \Delta x_{i_j}.$$

Lemma 8.11. For each $p > 0$ and bounded $f: [a, b] \rightarrow \mathbb{R}$, the set $B_p := \{x \in [a, b] : \omega_f(x) \geq p\}$ is compact.

Proof. Let $x_n \rightarrow x$ be a sequence in B_p and $0 < \varepsilon < p$. For each x_n , there is y_n such that $|f(x_n) - f(y_n)| > \varepsilon$. Now $\omega_f(x) \geq p$ is clear, so B_p is closed (and bounded). 

Theorem 8.12 (Lebesgue's Criterion for Riemann Integrability). The bounded function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable on $[a, b]$ iff f is continuous a.e. on $[a, b]$.

Proof. Let $\varepsilon > 0$ and f be continuous a.e. Pick a cover $\{(a_j, b_j)\}_1^{n-1}$ of $\mathcal{D}_f$, satisfying $\lambda\left(\bigcup_{j=1}^{n-1} (a_j, b_j)\right) < \frac{\varepsilon}{2\mathcal{M}(b-a)}$, where $\mathcal{M} := \sup f - \inf f$. Further define $b_0 := a$ and $a_n := b$. By continuity, for each $1 \leq j \leq n$ there is a partition $P_j = \{x_{i,j}\}_0^{n_j}$, such that

$$U(f|_{[b_{i-1}, a_i]}, P_i) - L(f|_{[b_{i-1}, a_i]}, P_i) < \frac{\varepsilon}{2n}.$$

Consequently, for $P := \{x_i\}_1^N := \{P_1 < P_2 < \cdots < P_n\}$, we have

$$U(f, P) - L(f, P) \leq \sum_{i=1}^n U(f|_{[b_{i-1}, a_i]}, P_i) - L(f|_{[b_{i-1}, a_i]}, P_i) + \sum_{j=1}^{n-1} \mathcal{M}(b_j - a_j) < \varepsilon.$$

The converse is just Lemma 8.10.

× Oops, at the start, I mixed up $[a, b]$ and $\mathcal{D}_f$ in my head.



Proof. Let $\varepsilon > 0$ and f be continuous a.e. Using 8.11, pick a cover $\{(a_j, b_j)\}_1^{n-1}$ of $B_{\varepsilon(b-a)^{-1/2}}$ satisfying $\lambda\left(\bigcup_{j=1}^{n-1} (a_j, b_j)\right) < \frac{\varepsilon}{2\mathcal{M}(b-a)}$, where $\mathcal{M} := \sup f - \inf f$. Further define $b_0 := a$ and $a_n := b$. By 8.9, for each $1 \leq i \leq n$, there is a partition P_i such that

$$U(f|_{[b_{i-1}, a_i]}, P_i) - L(f|_{[b_{i-1}, a_i]}, P_i) < \frac{\varepsilon}{2} \cdot \frac{b_{i-1} - a_i}{b - a}.$$

Consequently, for $P := \{P_1 < P_2 < \cdots < P_n\}$, we have

$$U(f, P) - L(f, P) \leq \sum_{i=1}^n U(f|_{[b_{i-1}, a_i]}, P_i) - L(f|_{[b_{i-1}, a_i]}, P_i) + \sum_{j=1}^{n-1} \mathcal{M}(b_j - a_j) < \varepsilon.$$

The converse is just 8.10.



Exercise 8.8. The oscillation as a measure of “jumps” in the function.

- Let $f(x) := \mathbf{1}_{[0, \infty)}$. Prove that $\omega_f(0) = 1$.
- Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded and let $x \in (a, b)$ be so that $\lim_{z \rightarrow x^+} f(z)$ and $\lim_{z \rightarrow x^-} f(z)$ both exist. Prove that $\omega_f(x) \geq |\lim_{z \rightarrow x^+} f(z) - \lim_{z \rightarrow x^-} f(z)|$.
- State when the equality in exercise 8.8(b) actually is an equality and prove your claim.

Proof.

(a)–(b) Trivial.

- We see that $\omega_f(x) = |\lim_{z \rightarrow x^+} f(z) - \lim_{z \rightarrow x^-} f(z)|$ iff $f(x)$ is between $\lim_{z \rightarrow x^-} f(z)$ and $\lim_{z \rightarrow x^+} f(z)$, inclusive.



Exercise 8.10. Prove that for $\mathbf{1}_{(-\infty, 0)}$ we have $\inf\{\omega_f(a, b) \mid a \leq 0 \leq b\} = 0$, even though f is discontinuous at zero. Then explain why this does not contradict exercise 8.8(b).

Proof. Since $\omega_f([0, 0]) = 0$, we obtain $\inf\{\omega_f(a, b) \mid a \leq 0 \leq b\} = 0$. Notice that the oscillation of f at a point only concerns itself with open intervals; $\omega_f(0) = 1$ in our case. Hence, no contradiction arises with 8.8(b).



Exercise 8.12. Let $a < b$. A function $f: [a, b] \rightarrow \mathbb{R}$ is said to be of *bounded variation* iff $V_a^b f := \sup V_a^b(f, P) < \infty$, where $V_a^b(f, P) := \sum_{i=1}^n |f(a_i) - f(a_{i-1})|$ for each partition $P = \{a = a_0 < a_1 < \cdots < a_n = b\}$ of $[a, b]$.

- (a) Prove that every function of bounded variation is the difference of two nondecreasing functions.
- (b) Prove that every function of bounded variation has at most countably many discontinuities.
- (c) Prove that every function of bounded variation on $[a, b]$ is Riemann integrable on $[a, b]$.
- (d) Prove that if f and g are of bounded variation and $\alpha \in \mathbb{R}$, then $f + g$ and αf are of bounded variation, too.

Proof.

- (a) Consider $g := \sup g_P$ where $g_P: [a, b] \rightarrow \mathbb{R}$ is given by $g_P(x) := f(a) + \sum_{a_i \leq x} |f(a_i) - f(a_{i-1})|$, for each partition $P = \{a_i\}_1^n$ of $[a, b]$. Pick partitions P and Q of $[a, b]$ and $[x, y]$, such that

$$g(x) - g_P(x) < \frac{\varepsilon}{3} \quad \text{and} \quad g(y) - g_P(y) < \frac{\varepsilon}{3} \quad \text{and} \quad V_x^y f - V_x^y(f, Q) < \frac{\varepsilon}{3}.$$

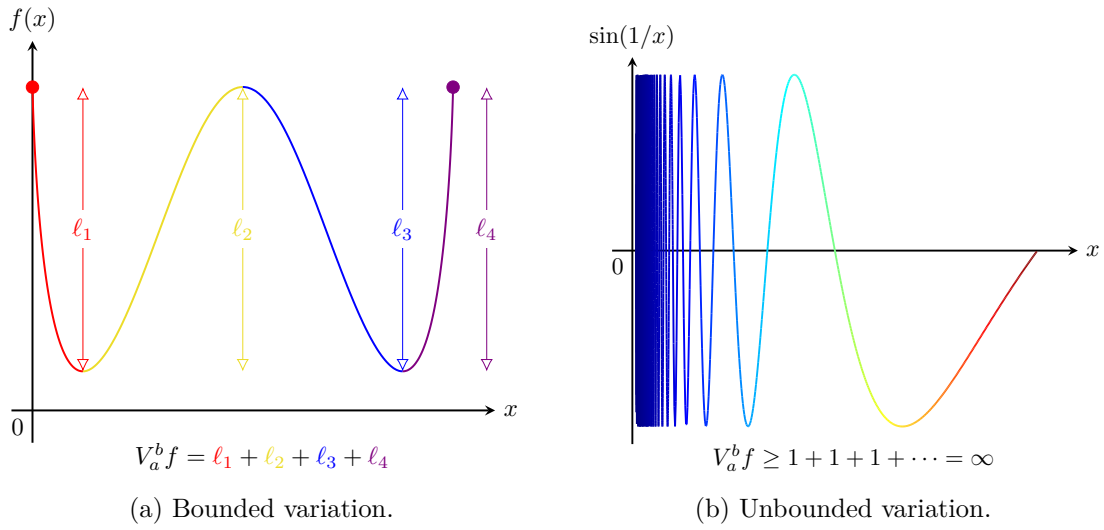
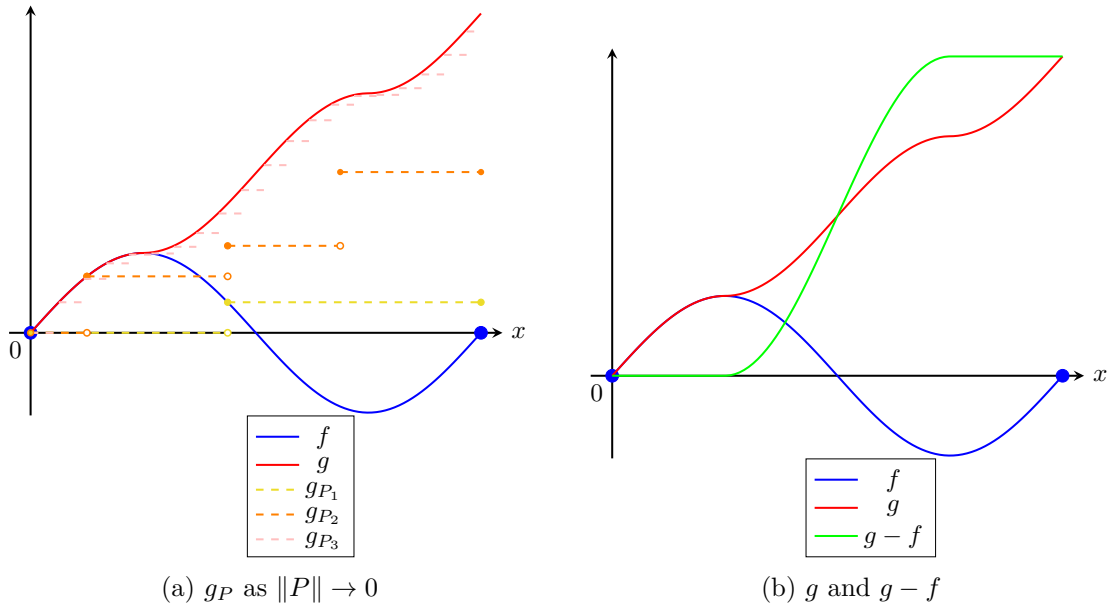
By the triangle inequality, for $R := P \cup Q$ we have

$$\begin{aligned} |V_x^y f - (g(y) - g(x))| &\leq V_x^y f - V_x^y(f, R \cap [x, y]) \\ &\quad + g(y) - g_R(y) + g(x) - g_R(x) \\ &\leq V_x^y f - V_x^y(f, Q) \\ &\quad + g(y) - g_P(y) + g(x) - g_P(x) \\ &< \varepsilon. \end{aligned}$$

i.e. $g(y) - g(x) = V_x^y f$. It follows that g and $g - f$ are nondecreasing. For an illustration, see Figure 8.6.

- (b)–(c) Since $\mathcal{D}_f \subseteq \mathcal{D}_g + \mathcal{D}_{g-f}$ is countable by 7.23, satisfying [Lebesgue's Criterion](#), the Riemann integrability of f is certain.
- (d) The triangle inequality makes the former clear: $V_a^b(f+g) \leq V_a^b f + V_a^b g < \infty$. Similarly, observe that $V_a^b(\alpha f) = |\alpha| V_a^b f < \infty$.



Figure 8.5: An illustration of the variation $V_a^b f$ of a function $f: [a, b] \rightarrow \mathbb{R}$.Figure 8.6: The case $f(x) = \sin(x)$.

Claim. Let $f: [a, b] \rightarrow \mathbb{R}$ and $c \in [a, b]$. Then, $V_a^b f = V_a^c f + V_c^b f$.

Proof. Let $\{p_i\}_1^n$ be a partition of $[a, b]$ and

$$\{q_i\}_1^m = \{p_1 < p_2 < \cdots < p_j < c < p_{j+1} < p_{j+2} < \cdots < p_n\}.$$

Then,

$$\sum_1^n |f(p_i) - f(p_{i-1})| \leq \sum_1^m |f(q_i) - f(q_{i-1})| \leq V_a^c f + V_c^b f$$

by the triangle inequality. So, $V_a^b f \leq V_a^c f + V_c^b f$. Furthermore, since $R := \{P < Q\}$ is a partition of $[a, b]$ — for any partitions P of $[a, c]$ and Q of $[c, b]$ — it is clear that $V_a^b f \geq V_a^c f + V_c^b f$.



Question. Let $f: [a, b] \rightarrow \mathbb{R}$ have unbounded variation. Does there exist $x \in [a, b]$ such that $f|_{[c, d]}$ is also of unbounded variation, for all $c < x < d$?

Question. So, we know the variation of a function is (finitely) additive, in the above sense. But, is it countably additive? Countably subadditive?

Proof. Suppose this is false, for contradiction — for each x , we find $c_x < x < d_x$ such that $f|_{[c_x, d_x]}$ has bounded variation. By compactness, there is a finite subcover $\{[c_{x_n}, d_{x_n}]\}_1^N$ of $[a, b]$. Hence,

$$V_a^b f = \sum_{n=1}^N V_a^b f|_{[c_{x_n}, d_{x_n}]} < \infty.$$



Remark. Courtesy of discontinuities, in general we have

$$V_a^b(f, P) \not\xrightarrow{\|P\| \rightarrow 0} V_a^b f \quad \text{and} \quad g_P \not\xrightarrow{\|P\| \rightarrow 0} g.$$

Exercise 8.14. Prove that if $A, N \subseteq \mathbb{R}$ and N is a null set, then $\lambda(A - N) = \lambda(A)$.

Proof. This is clear from

$$\lambda(A) = \lambda((A - N) \cup N) \leq \lambda(A - N).$$



Exercise 8.15. Prove that if C^Q is a *Cantor set* with nonzero measure (see exercise 8.3(e)), then $\mathbf{1}_{C^Q}$ is not Riemann integrable.

Proof. Let $x \in C^Q$; for each n , there is a unique i_n such that $x \in I_{i_n, n}^Q$. So, pick

$$x_{n+1} \in I_{i_n, n}^Q - \left(L_{q_{n+1}}[I_{i_n, n}^Q] \cup R_{q_{n+1}}[I_{i_n, n}^Q] \right).$$

Since $\mathbf{1}_{C^Q}(x_n) = 0 \rightarrow 0 \neq 1$, we conclude $C^Q \subseteq \mathcal{D}(\mathbf{1}_{C^Q})$ is not a null set.



8.3 More Integral Theorems

Theorem 8.17 (Fundamental Theorem of Calculus, Derivative Form). Let f be a Riemann integrable function on $[a, b]$. Then the function $G(x) := \int_a^b f(t) dt$ is (uniformly) continuous on $[a, b]$. Furthermore, if f is continuous at $x \in (a, b)$, then G is differentiable at x with $G'(x) = \frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$.

Proof. Let $\varepsilon > 0$ and $y \in N_{\frac{\varepsilon}{\omega_f}}(x)$. Since $|\int_x^y f(t) dt| < \varepsilon$, the (uniform) continuity of G holds. Now pick $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$, for all $y \in N_\delta(x)$. So,

$$\left| \int_y^x f(t) dt - (x - y)f(x) \right| \leq U(f, \{x, y\}) - L(f, \{x, y\}) < \varepsilon|x - y|.$$

i.e. G is differentiable at x with $G'(x) = f(x)$.

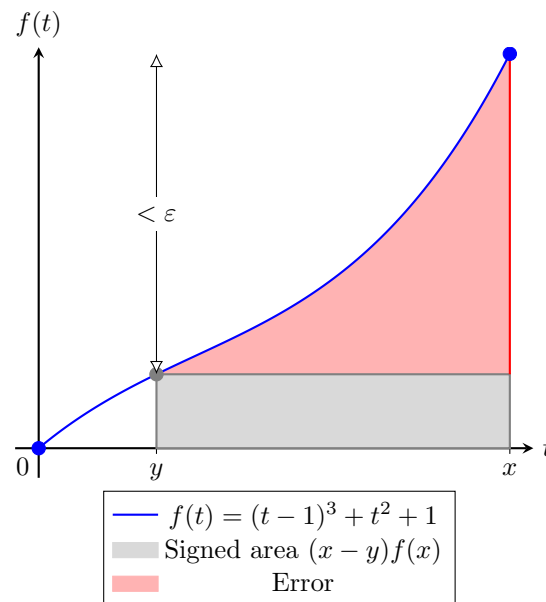


Figure 8.7: An illustration of theorem 8.17. Notice that $\text{Area}(\triangle) < \varepsilon|x - y|$.

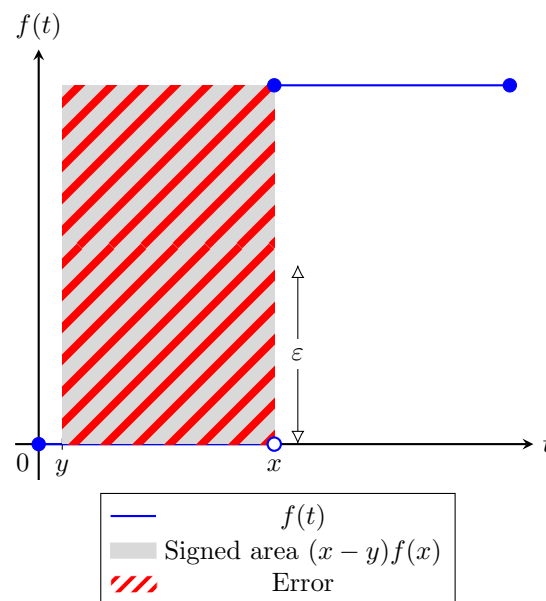


Figure 8.8: An illustration of how theorem 8.17 fails for jump discontinuities at x .

Claim. Let f be Riemann integrable on $[a, b]$ and $G(x) := \int_a^x f(t) dt$. Then, the limit $\lim_{t \rightarrow x} f(t)$ exists iff G is differentiable at x . In fact, $G'(x) = \lim_{t \rightarrow x} f(t)$.

Proof. This is false. Consider

$$G(x) := \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin(1/x) & \text{if } x \neq 0, \end{cases}$$

$$f(x) := \begin{cases} 0 & \text{if } x = 0, \\ 2x \sin(1/x) - \cos(1/x) & \text{if } x \neq 0. \end{cases}$$

Then, $G(x) = \int_0^x f(t) dt$ by theorem 5.23; $G'(0) = 0$ even though $\lim_{t \rightarrow 0} f(t)$ does not exist. 

Note. If we consider the Lebesgue integral (instead of restricting ourselves to its Riemannian counterpart), then Dirichlet's function $\mathbf{1}_{\mathbb{Q}}$ provides an immediate counterexample — $G'(r) = 1 = \mathbf{1}_{\mathbb{Q}}(r)$ for rational r .

Exercise 8.20. Determine which result from this section that was used in the proof of theorem 8.17 was not available in section 5.3. (This prevented us from placing theorem 8.17 right after the Antiderivative Form of the Fundamental Theorem of Calculus.)

Proof. Theorem 8.16 — that a function Riemann integrable over $[a, b]$ is also Riemann integrable over any closed subinterval of $[a, b]$. 

Exercise 8.21 (Mean Value Theorem for the Integral). Let the function $f: [a, b] \rightarrow \mathbb{R}$ be continuous and let the function $g: [a, b] \rightarrow [0, \infty)$ be nonnegative and Riemann integrable. Prove there is a $c \in [a, b]$ so that the integral satisfies

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx.$$

Proof. Let $C := \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx}$, $m := \min f$ and $M := \max f$. We notice

$$\begin{array}{ccccc} m & \leq & f(x) & \leq & M \\ \int_a^b m g(x) dx & \leq & \int_a^b f(x)g(x) dx & \leq & \int_a^b M g(x) dx \\ m & \leq & C & \leq & M \end{array}$$

So, $C = f(c)$ for some $c \in [a, b]$, by the Intermediate Value Theorem (3.34). 

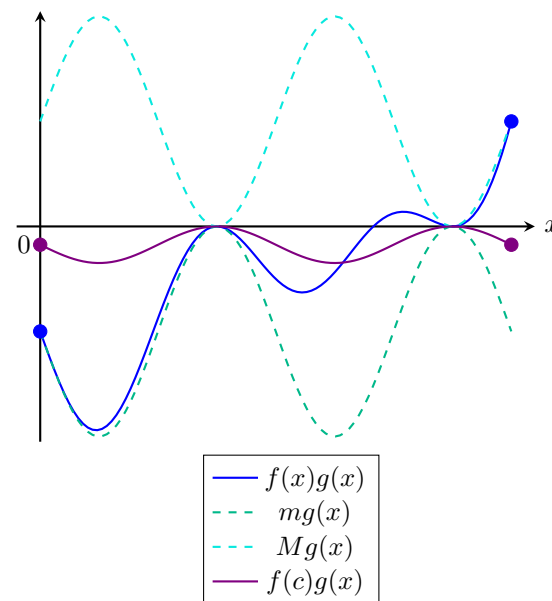
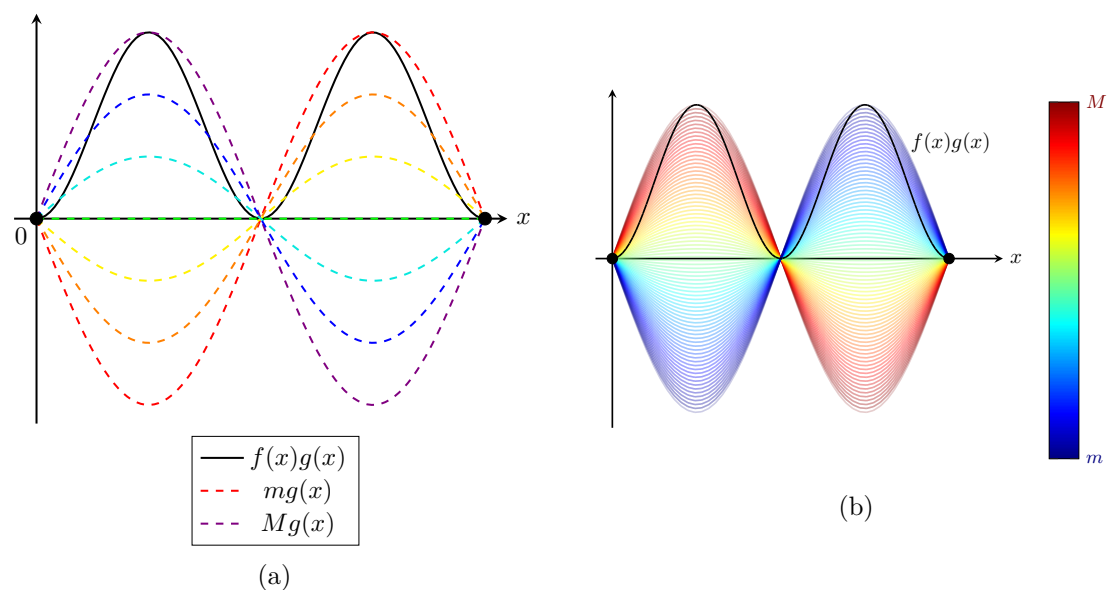


Figure 8.9: An illustration of theorem 8.21.

Figure 8.10: The importance of g being nonnegative/nonpositive: $\min\{mg, Mg\} \leq fg \leq \max\{mg, Mg\}$ is insufficient for the result of theorem 8.21 to hold. The above case of $f(x) = g(x) = \sin(x)$ provides an example.

Exercise 8.22. Let $g: [a, b] \rightarrow [0, \infty)$ be Riemann integrable and $f: [a, b] \rightarrow \mathbb{R}$ be nondecreasing. Prove that there is $c \in [a, b]$ so that

$$\int_a^b f(x)g(x) dx = f(a) \int_a^c g(x) dx + f(b) \int_c^b g(x) dx.$$

Proof. Since

$$f(a) \int_a^b g(x) dx \leq \int_a^b f(x)g(x) dx \leq f(b) \int_a^b g(x) dx,$$

the result follows from the continuity of the definite integral (theorem 8.17). 

Exercise 8.28. Construct a function $f: [0, 1] \rightarrow \mathbb{R}$ so that $|f|$ is Riemann integrable and f is not Riemann integrable.

Proof. Consider

$$f(x) := \begin{cases} -1 & \text{if } x \in \mathbb{Q}, \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$



Exercise 8.29. A function $f: [a, b] \rightarrow \mathbb{R}$ is called *absolutely continuous* iff for every $\varepsilon > 0$ there is $\delta > 0$ so that for all sequences $\{(a_i, b_i)\}_1^n$ of pairwise disjoint open intervals the inequality

$$\sum_{i=1}^n (b_i - a_i) < \delta \quad \text{implies} \quad \sum_{i=1}^n |f(b_i) - f(a_i)| < \varepsilon.$$

- (a) Let $f: [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function. Prove that the function $G: [a, b] \rightarrow \mathbb{R}$ defined by $G(x) := \int_a^x f(t) dt$ is absolutely continuous on $[a, b]$.
- (b) Prove that every absolutely continuous function is uniformly continuous.
- (c) Prove that $f(x) = 1/x$ is continuous, but not absolutely continuous on $(0, 1]$.

Proof.

- (a) If $\sum_{i=1}^n b_i - a_i < \frac{\varepsilon}{\omega_f}$, then

$$\sum_{i=1}^n |G(b_i) - G(a_i)| = \sum_{i=1}^n \left| \int_{a_i}^{b_i} f(t) dt \right| \leq \sum_{i=1}^n (b_i - a_i) \omega_f < \varepsilon.$$

- (b) Uniform continuity is merely the case of $n = 1$ in absolute continuity.
- (c) It is not uniformly continuous, let alone absolutely continuous: $f(x) - f(2x) = \frac{1}{2x} \rightarrow \infty$ when $2x - x = x \rightarrow 0$.



Remark. Essentially, (a) is saying that total area tends to zero, as the total base length tends to zero.

Question. Is the following equivalent to the continuity of $f: [a, b] \rightarrow \mathbb{R}$? For each $\varepsilon > 0$, there is $\delta > 0$ such that $V_x^{x+\delta} f < \varepsilon$.

Proof. Suppose, for contradiction, f is continuous but there is $\varepsilon > 0$, such that $V_x^{x+\delta} f < \varepsilon$ for all $\delta > 0$. Thus, pick 

8.4 Improper Riemann Integrals

Example 8.21. The *p*-integral test for integrals over infinite intervals. Let $p > 0$ be rational. Then $f(x) = 1/x^p$ is improperly Riemann integrable over $[1, \infty)$ iff $p > 1$.

Proof. If $p > 1$, then the Fundamental Theorem of Calculus implies

$$\int_1^x \frac{1}{t^p} dt = -\frac{1}{p-1} \left(\frac{1}{x^{p-1}} \right) \rightarrow \frac{1}{p-1} \quad \text{as } x \rightarrow \infty.$$

So, f is improperly Riemann integrable. Conversely, when $0 < p \leq 1$,

$$\int_1^x \frac{1}{t^p} dt \geq \sum_{n=1}^{\lfloor x \rfloor - 1} \frac{1}{(n+1)^p}.$$

Since $\left\{ \sum_{n=1}^{m-1} \frac{1}{(n+1)^p} \right\}_{m=1}^{\infty}$ is nondecreasing, the Ratio Test gives $\lim_{x \rightarrow \infty} \int_1^x \frac{1}{t^p} dt = \infty$. That is, f is not improperly Riemann integrable. 

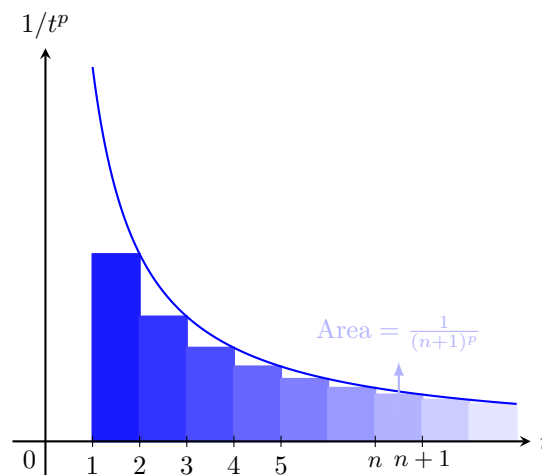


Figure 8.11

Proposition 8.23. Let $a, b \in \mathbb{R}$ with $a < b$ and let $f: [a, b] \rightarrow \mathbb{R}$ be Riemann integrable. Then the improper Riemann integral of f over $[a, b)$ exists and it is equal to the Riemann integral of f over $[a, b]$.

Proof. A corollary of theorem 8.17. 

Theorem 8.26. Let $f: [a, b) \rightarrow \mathbb{R}$ (where $b \in \overline{\mathbb{R}}$) be Riemann integrable over all $[a, c] \subseteq [a, b)$. Then f is improperly Riemann integrable over $[a, b)$ iff for all $c \in [a, b)$ the function f is improperly Riemann integrable over $[c, b)$ and in this case

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Proof. Suppose f is improperly Riemann integrable and let $c \in [a, b)$. Then,

$$\int_c^b f(x) dx = \lim_{t \rightarrow b^-} \left(\int_a^t f(x) dx - \int_a^c f(x) dx \right) = \int_a^b f(x) dx - \int_a^c f(x) dx,$$

proving the improper Riemann integrability of f over $[c, b)$. For the converse, select $c = a$. 

Theorem 8.27 (Comparison Test for improper integrals). Let $f: [a, b) \rightarrow \mathbb{R}$ be Riemann integrable over every closed interval in $[a, b)$ and $g: [a, b) \rightarrow \mathbb{R}$ be improperly Riemann integrable over $[a, b)$, such that $0 \leq f \leq g$. Then f is improperly Riemann integrable over $[a, b)$, such that

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

Proof. Since $\int_a^t f(x) dx$ is nondecreasing and bounded, $\int_a^b f(x) dx$ must be just $\sup_{t \in [a, b)} \int_a^t f(x) dx$. So, f is improperly Riemann integrable over $[a, b)$. The final inequality is implied by limit laws. 

Question. Let $f: [a, b) \rightarrow \mathbb{R}$ be Riemann integrable over every closed interval in $[a, b)$ and $g: [a, b) \rightarrow \mathbb{R}$ be improperly Riemann integrable over $[a, b)$, such that $-g \leq f \leq g$. Then must f be improperly Riemann integrable over $[a, b)$, such that

$$\left| \int_a^b f(x) dx \right| \leq \left| \int_a^b g(x) dx \right| ?$$

Theorem 8.28. Let $f: [a, b) \rightarrow \mathbb{R}$ be Riemann integrable over every closed interval in $[a, b)$. If $|f|$ is improperly Riemann integrable over $[a, b)$, then f is improperly Riemann integrable over $[a, b)$ and the *triangular inequality*

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

holds.

Proof. Suppose $|f|$ is improperly Riemann integrable over $[a, b)$. Since $0 \leq f(x) - \min f \leq |f(x)| - \min f$, theorem 8.27 implies the improper Riemann integrability of f over $[a, b)$. Lastly, the required *triangular inequality* is obtained from theorem 8.14. 

Question. Is there a criterion for improper Riemann integrability, that parallels that of [Lebesgue's Criterion for Riemann Integrability](#)?

- (a) Is there some function $\Delta: [a, b) \times [a, b) \rightarrow \mathbb{R}$, such that any function $f: [a, b) \rightarrow \mathbb{R}$ is Riemann integrable iff f is continuous a.e. and $|f(x) - f(y)| \leq \Delta(x, y)$

a.e.

(b) Let the signed variation U_a^b of a function $f: [a, b] \rightarrow \mathbb{R}$ be given by

$$U_a^b f :=$$

Is there some function $u: [a, b] \times [a, b] \rightarrow \mathbb{R}$, such that any function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff f is continuous a.e. and $\frac{U_y^x f}{x-y} \leq u(x, y)$ a.e.?

(work in progress question statement)

Proof.

(a) No. Suppose



Theorem 8.29 (Integral Test). Let $f: [1, \infty) \rightarrow [0, \infty)$ be a bounded nonincreasing function. Then the series $\sum f(n)$ converges iff the improper integral $\int_1^\infty f(x) dx$ converges.

Proof. First suppose $\sum f(n)$ converges. Recall that since f is nonincreasing, it is Riemann integrable over all closed subintervals of $[1, \infty)$. Furthermore, $0 \leq f \leq \sum f(n) \mathbf{1}_{[n, n+1)}$ so $\int_1^\infty f(x) dx$ converges by theorem 8.27.

Now consider when $\int_1^\infty f(x) dx$ converges. Then, as $\left\{ \sum_{n=1}^N f(n) \right\}_{N=1}^\infty$ is a non-decreasing sequence bounded by $f(1) + \int_1^\infty f(x) dx$, the series $\sum f(n)$ must converge.

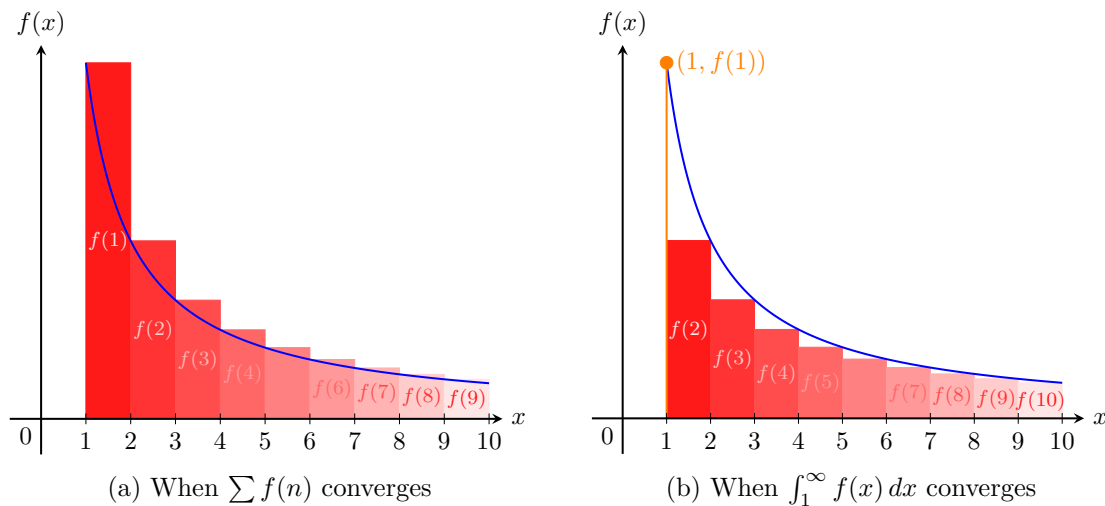


Figure 8.12: An illustration of theorem 8.29.

Note 8.30. The importance of f being nonincreasing.

($\Rightarrow$) Though $\sum |\sin(n\pi)| = 0$, we have $\int_1^\infty |\sin(\pi x)| dx = \infty$.

($\Leftarrow$) Although $\int_1^\infty \mathbf{1}_{\mathbb{N}} dx = 0$, we see that $\sum \mathbf{1}_{\mathbb{N}}(n) = \infty$.

Exercise 8.38. Construct a function $f: [1, \infty) \rightarrow [0, 1]$ that is improperly Riemann integrable, but does not converge to zero as $x \rightarrow \infty$.

Proof. Consider

$$f(x) := \begin{cases} x & \text{if } x \in \mathbb{N}, \\ 1/x^2 & \text{if } x \notin \mathbb{N}. \end{cases}$$

Over closed intervals, f and $1/x^2$ disagree on only a finite set. It is hence clear, that $\int_1^t f(x) dx = \int_1^t 1/x^2 dx$ converges as $t \rightarrow \infty$. 

Alternatively, using the *hint* — triangles of height 1 and area $1/2^n$ — obtain the following example, wherein $f(x)$ converges as $x \rightarrow \infty$.

Proof (Hint). Let

$$f(x) := \begin{cases} 2^n \left(x - \sum_{i=1}^n \frac{1}{2^i} \right) & \text{if } x \in \left(\sum_{i=1}^n \frac{1}{2^i}, \sum_{i=1}^n \frac{1}{2^i} + \frac{1}{2^{n+2}} \right) \\ & \text{for some } n \in \mathbb{N}, \\ -2^n \left(x - \sum_{i=1}^n \frac{1}{2^i} - \frac{1}{2^{n+2}} \right) & \text{if } x \in \left(\sum_{i=1}^n \frac{1}{2^i} + \frac{1}{2^{n+2}}, \sum_{i=1}^{n+1} \frac{1}{2^i} \right) \\ & \text{for some } n \in \mathbb{N}. \end{cases}$$

Obtuse formalism aside, f just comprises of triangles of height 1 and width/area $1/2^n$. So, though $\lim_{x \rightarrow \infty} f(x) = 1$,

$$\int_1^\infty f(x) dx = \sum \frac{1}{2^n} = 1.$$



Claim. Suppose $f, g: [a, b] \rightarrow \mathbb{R}$ agree a.e. Then, $f \in \mathcal{R}$ iff $g \in \mathcal{R}$; in fact, $\int_a^b f(x) dx = \int_a^b g(x) dx$.

Exercise 8.39 (Limit Comparison Test for improper integrals). Let $f, g: [a, b) \rightarrow [0, \infty)$ be Riemann integrable over all intervals $[a, c] \subseteq [a, b)$. Prove that if $\lim_{x \rightarrow b^-} f(x)/g(x) = K > 0$, then $\int_a^b f(x) dx$ converges iff $\int_a^b g(x) dx$ converges.

Proof. Suppose $\int_a^b g(x) dx$ converges and let $0 < \varepsilon < K$. Pick a corresponding $M \in [a, b)$, such that $0 < (K - \varepsilon)g(x) < f(x) < (K + \varepsilon)g(x)$, for each $x \in (M, b)$. Now, theorem 8.27 implies the improper Riemann integrability of f over $[a, b)$. For the converse, notice $\lim_{x \rightarrow b^-} g(x)/f(x) = 1/K > 0$. 

Exercise 8.40. Prove that the function $f: (a, b] \rightarrow \mathbb{R}$ is improperly Riemann integrable over $(a, b]$ iff the function $g(x) := f(a + 1/x) \cdot 1/x^2$ is improperly Riemann integrable over $\left[\frac{1}{b-a}, \infty\right)$ and that in this case the integrals are equal.

Proof. We see that

$$\int_{a+1/t}^b f(x) dx = \int_{b-a}^{1/t} u^2 f(a+u) \left(-\frac{1}{u^2}\right) du = \int_{\frac{1}{b-a}}^t \frac{1}{x^2} f\left(a + \frac{1}{x}\right) dx.$$

The result is hence apparent.



Resources

9.1 Chapter 4: Differentiable Functions

- (a) Information on discontinuous derivatives
 - i. <https://math.stackexchange.com/questions/292275/discontinuous-derivative>
 - ii. <https://math.stackexchange.com/questions/112067/how-discontinuous-can-a-derivative-be>